Keep It or Close It?
A Pareto-Improving Transition for Social Security

HongSeok Kim*

Job Market Paper

November 2026

Link to the latest version

Abstract

A smaller pay-as-you-go pension raises output in the long run, but the generations that paid the full payroll tax and then retire on reduced benefits are left worse off. I ask whether Social Security can be made sustainable without any losing generation, in an overlapping-generations model of the U.S. economy in which work builds skills and the government borrows up to a debt ceiling and closes the rest with a consumption tax. Against a baseline that borrows until debt reaches 150% of GDP and then pays what payroll revenue covers, lowering the payroll tax from 12.4% to 3.5% leaves none of birth cohorts worse off on average once two cohort-specific instruments, a lower payroll rate while working and a higher benefit late in retirement, are financed from general revenue. They cost 6% of 2026 GDP in present value. The gains come from more work, faster skill accumulation, and more saving under the lower payroll tax. The retained program yields higher social welfare than a complete phase-out, which leaves 51 cohorts worse off, for social discount factors up to 0.87. The same two instruments make the transition Pareto improving at every long-run payroll rate from zero to 6.5%, so how large a program to keep is not an efficiency question; it depends on how much weight future generations receive. Removing the bequest motive does not change these conclusions.

Keywords: Social Security reform, Pareto-improving transition, intergenerational redistribution, debt ceiling, learning by doing.

JEL: H55, H63, E62, D15, J26.

*I am deeply grateful to my advisor, Juan Carlos Córdoba, for invaluable advice, guidance, and encouragement. I am grateful to my dissertation committee, Rajesh Singh, Quinn Weninger, Sergio Lence, and Jian Li, for their continued support. I also thank Kathleen McGarry, John Winters, Agnitra Roy Choudhury, Juan Sánchez and Xiaobo Yu for thoughtful comments and feedback, as well as participants at the 2026 North American Summer Meeting of the Econometric Society, the 2026 WEAI Annual Conference, the EGSC at Washington University in St. Louis, and the Iowa State Micro/Macro Reading Group. Department of Economics, Iowa State University, Email: hongseok@iastate.edu.

1 Introduction

In 2034 the Social Security trust fund is projected to be depleted,[1] and the payroll tax that remains will cover about 83% of the benefits that current law promises (Social Security Trustees, 2026). Congress will then have to raise taxes, cut benefits, or borrow, and each of these choices has the same problem: someone alive today pays for it, either in higher taxes during their working years or in smaller checks during retirement. Is that unavoidable? This paper asks whether Social Security can be reformed so that no generation, present or future, ends up worse off on average than under a current law, and whether it is better to keep a smaller program or to close it altogether.

Literature has studied this question for decades, and the answer has mostly been no. A pay-as-you-go pension gives its first generation benefits it never paid for, and later generations earn a return equal to wage growth rather than the interest rate (Samuelson, 1958; Aaron, 1966). That initial windfall is a debt that someone must eventually carry, so moving toward a funded system cannot avoid making some generation worse off (Breyer, 1989; Geanakoplos et al., 1998; Sinn, 2000). Quantitative life-cycle models confirm this along the transition: future cohorts gain and the cohorts alive at the reform lose (Auerbach and Kotlikoff, 1987; Huang et al., 1997; Conesa and Krueger, 1999; Nishiyama and Smetters, 2007). The exception is McGrattan and Prescott (2017), who find a phase-out of Social Security that leaves every cohort better off, but only as part of a broader tax reform and in an economy without the human capital over working lives that they identify as the missing ingredient. What remains open is whether a program that is kept, not closed, can be made sustainable without any losing cohort, using only instruments that operate within Social Security itself, and when keeping it is better than closing it.

I study this question in a large-scale overlapping-generations model in the tradition of Auerbach and Kotlikoff (1987), with the demographic and fiscal structure of Braun and Joines (2015): households enter at 20, face mortality risk, retire at 65 and live to at most 89, and the population ages along the Trustees' projected path. Households differ in assets and skills, value the estates they leave (De Nardi, 2004), and build skills through work (Imai and Keane, 2004; Chang et al., 2002), so the taxes that finance pensions change future earnings as well as current hours. The government runs Social Security and the Treasury as separate accounts, borrows up to a debt ceiling, and closes the remaining gap with a consumption tax, the financing margin used in Kitao (2014) and Braun and Joines (2015).

[1]The OASI and DI trust funds are legally separate. The combined projection assumes legislation that permits transfers between them. The calibration uses this combined projection.

I discipline the model so that it reproduces the features of the U.S. economy that decide how a pension reform plays out. Households save, work, and build skills at the rates observed in 2010: the economy holds three years of output in capital, adults work the hours recorded in the American Time Use Survey, and wages rise and then flatten over the working life as they do in the Current Population Survey, so that the response of saving, labor supply, and future earnings to a tax or benefit change is anchored to data rather than assumed. Wealth is as unequal within each age group as in the Survey of Consumer Finances, and the flow of estates matches what the wealth and mortality data imply, so the reform's effect on who inherits and who leaves nothing is also disciplined. On the fiscal side, the model starts from the observed public debt and trust-fund reserves, Social Security's benefits in payment and benefit formula, its payroll base, and government spending from the national accounts. The population then ages along the Trustees' projection, and the model reproduces the Trustees' reserve-exhaustion year and their projected program cost in 2100.

The transition path follows from the economics of the problem. In the baseline, benefits fall to what payroll revenue covers once the government can no longer borrow. Shrinking the program instead creates resources: a lower payroll tax raises the return to work, more work builds skills and future earnings, and smaller benefits lead households to save more, so capital, wages, and output rise. These gains accrue mostly to cohorts who spend their working lives under the lower tax. The cohorts who lose are those who pay the full payroll tax for most of their careers and then retire into the reduced benefits. A reform that leaves no one worse off must therefore move part of the gain to these cohorts, and how it does so matters: if it finances the transfer with a consumption tax, the same retirees it means to protect pay part of the bill.

This shapes the reform design. I announce the cut early but start it late enough that today's retirees and older workers are untouched, and I cut gradually so that each cohort's benefit loss is matched by payroll savings over its own working years. The cohorts caught in between, who pay the old tax for part of their careers and retire into the lower benefits, I compensate on their own margins: a lower payroll rate while they work and a higher benefit after they retire. The supplement is paid late in life, so its cost falls in years when most taxpayers are themselves beneficiaries of the reform. General revenue pays for both instruments, so no single cohort is singled out to finance another. Public debt stays at its ceiling throughout, so the compensation is paid out of current revenue, which the higher output makes possible, rather than borrowed from the future. I measure each cohort's welfare as the population-weighted average of its members' lifetime utility and size the two

instruments cohort by cohort until no birth cohort loses. To rank reforms, I aggregate these cohort welfares with a social discount factor over those not yet born. I then compare the retained program with a complete phase-out and with versions of the retained program at other long-run payroll rates, debt ceilings, and financing rules, and I repeat the main comparison in a recalibrated economy without a bequest motive.

None of the pieces of this reform is new in practice. Germany, Japan, Canada, and Australia all pay part or all of their public pensions from general revenue, and so does the United States for Medicare Parts B and D (Medicare Trustees, 2025). Pension reforms are routinely phased in by birth cohort, as the 1983 Social Security amendments did with the retirement age, and Chile compensated the cohorts caught in its 1981 transition with Treasury-paid recognition bonds. Since 2022 Canada pays Old Age Security recipients aged 75 and over a benefit ten percent higher than younger recipients receive. This paper combines these instruments into one transition and asks how large each must be, cohort by cohort, for no generation to lose, and whether a government that holds its debt at a fixed ceiling can afford them.

The main finding is that such a transition exists and that it is not costly. Cutting the payroll tax from 12.4% to 3.5% between 2060 and 2200, with the two compensation instruments sized cohort by cohort, leaves none of the birth cohorts I evaluate worse off on average than under the baseline. The compensation is small relative to the gain it protects. The largest payroll relief lowers a cohort's payroll tax rate by 2.3 percentage points below the economy-wide rate, from 12.4% to 10.1%. The largest late-life supplement raises a cohort's annual benefit by 8.6% above what it would otherwise receive. Summed over the transition and discounted, the Treasury's spending on the two instruments equals 6% of GDP in 2026. Because public debt stays at its ceiling, the consumption tax must raise this money. The consumption tax rate exceeds the baseline's by at most 0.45 percentage points while the relief is phased in, and from 2072 on it is lower, because output is larger while government spending outside Social Security grows with productivity rather than with output, and the interest rate on the public debt is lower. By 2200, capital per adult and consumption per adult are 23%, 16% higher than in the baseline, respectively. A cohort born in 2100 gains 3.4% of lifetime consumption.

The retained program also delivers higher social welfare than a complete phase-out for every social discount factor up to 0.87, and a planner who counts only people alive today records the phase-out as a net loss. The phase-out produces larger gains in the long run, but it reaches them by cutting benefits in 2033 rather than 2060 and by compensating no one, the

cohort born in 1951 loses the equivalent of 5.5% of its consumption in every remaining year of life. Only a planner who weights the unborn heavily prefers the phase-out: at a discount factor of 0.97, its gain in social welfare, in lifetime-utility units per adult alive in 2026, is five times that of the retained program. I set the long-run payroll rate at 3.5% because, among the rates I solve, it gives the highest social welfare for social discount factors between 0.83 and 0.91. The rate matters less than the timing and the compensation: re-solving the compensation at rates from zero to 6.5% makes every one of them Pareto improving with the same two instruments, and the welfare differences across rates stay within a few hundredths of a utility unit except for a planner who weights the unborn heavily, for whom lower rates do better.

Section 6 checks the result along four dimensions, and in each case the compensation is re-solved so that no cohort loses. Removing the bequest motive, with the assets of the deceased passed to the survivors of their own cohort and the economy recalibrated, leaves the Pareto result and the comparison with the phase-out intact. Keeping the retirement consumption floor in place changes nothing, because no retiree reaches it. Financing the payroll relief with a surcharge on cohorts born from 2044 onward, rather than from general revenue, lowers social welfare only slightly. Letting debt rise a few points above the ceiling while the compensation is paid raises it slightly, and stretching the cut to 2300 gives up most of the gain. The ranking of the retained program and the phase-out is the same in every case.

Related Literature. The paper belongs to the literature on the transition cost of pension reform. Breyer (1989), Geanakoplos et al. (1998), and Sinn (2000) show that the windfall paid to the first generation of a pay-as-you-go system is an implicit debt that a move toward funding cannot erase, so some generation must carry it. Quantitative life-cycle models confirm the pattern along the transition: in Auerbach and Kotlikoff (1987), Huang et al. (1997), Conesa and Krueger (1999), and Nishiyama and Smetters (2007), future cohorts gain from privatization and the cohorts alive at the reform lose, and a Pareto improvement requires lump-sum transfers across cohorts of the kind a legislature cannot write. Kitao (2014) compares the financing options, higher payroll taxes, lower benefits, a later retirement age, and a consumption tax, under a fixed fiscal rule, and Braun and Joines (2015) and İmrohoroğlu et al. (2016) place the problem in an aging economy with Medicare and public debt. McGrattan and Prescott (2017) come closest to this paper: in an economy with annuities and no skill formation, they find a phase-out of Social Security and Medicare that leaves every cohort better off, but only together with a broader income-tax reform. I keep their question and change three things. The compensation uses instruments inside

Social Security and the general budget, cohort-specific payroll relief and a late-life benefit supplement, rather than a reform of the tax base. The program is retained at a positive long-run size, and the terminal-rate sweep shows that the Pareto property is a feature of the transition design rather than of the destination. And work builds skills, the channel their paper names as omitted, so that the taxes that finance pensions change future earnings as well as current hours.

A second literature asks what size of Social Security is optimal. Positive answers rest on insurance: against longevity when annuity markets are missing, against earnings and health risk, and through redistribution across skill types (İmrohoroğlu et al., 1995; Hubbard et al., 1995; Huggett and Ventura, 1999; Conesa and Krueger, 1999; Hosseini and Shourideh, 2019), with Conesa and Garriga (2008) and Peterman and Sager (2022) studying the optimal mix of social security, taxes, and public debt in life-cycle economies. Earnings and health are deterministic here, and the estate pool together with the bequest motive leaves little longevity risk to insure, so the long-run optimum in this economy is zero, as the Aaron condition predicts when the interest rate exceeds the growth rate (Aaron, 1966). What the paper isolates is the transition component of the optimal size: with a no-losers constraint and implementable instruments, the constrained-optimal landing rate is positive over a range of social discount factors because a shallower cut leaves more of the cohorts born between 2007 and 2035 with gains of their own instead of compensation. The insurance component would add to that level.

The model's labor-supply and fiscal blocks follow established work. Learning by doing, as in Imai and Keane (2004), Chang et al. (2002), and Wallenius (2011), links current hours to future earnings and the tax base; Blandin (2021) and Peterman (2016) show that endogenous human capital changes the welfare effects of payroll and income taxes in the same direction found here. Warm-glow bequests follow De Nardi (2004), the means-tested consumption floor Hubbard et al. (1995) and De Nardi et al. (2010), and the demographic and medical blocks Attanasio et al. (2011) and Braun and Joines (2015). The fiscal rule, a debt ceiling enforced by a consumption tax, follows Bohn (1998) and the aging-and-debt models of Altig et al. (2001), Hansen and İmrohoroğlu (2016), and İmrohoroğlu et al. (2016). Because the interest rate exceeds the growth rate throughout, borrowing is not free in this economy, in contrast to the low-rate settings of Blanchard (2019), Eggertsson et al. (2019), and Auclert et al. (2021); the ceiling is the device that forces the compensation to be paid out of the efficiency gain rather than passed to future taxpayers. Finally, the design responds to the political-economy literature's concern that reforms create losers who block them (Tabellini, 1991; Bouton et al.,

2020; Diamond, 1997): cohort-specific rules of the kind used here, phased retirement ages and recognition bonds, are the instruments actual reforms have used (Holzmann et al., 2004).

Section 2 presents the model and Section 3 its calibration. Section 4 constructs the transition and reports its welfare and macroeconomic effects. Section 5 compares the retained program with a complete phase-out and across long-run payroll rates. Section 6 examines the bequest motive, the financing of the compensation, and the borrowing allowance. Section 7 concludes.

2 Model

The model is a general-equilibrium overlapping-generations economy based on Auerbach and Kotlikoff (1987). Its demographic and fiscal components are similar to those in Braun and Joines (2015). Each model period is one year.

Households enter the model at age 20. They retire at age 65 and die by age 90. Households differ in their assets and skills. They choose consumption, saving, and hours of work. They cannot borrow. Their skills increase through work. Households also value the bequests they leave. Mortality is their only source of idiosyncratic risk. Firms use capital and efficiency units of labor to produce one good. The government has four components. Social Security pays benefits from payroll taxes and its trust fund. Medicare and a social safety net provide support for the elderly. The Treasury uses a consumption tax to balance the overall government budget. Households have perfect foresight along deterministic policy paths.

2.1 Demographics and productivity

Let $N_{j,t}$ be the population of age j at the start of year t. Let $\pi_{j,t}$ be the probability that a person of age j survives year t. Everyone of the same age has the same survival probability. Both series come from the intermediate projection in the 2026 Trustees Report. The projection gives data by single year of age and sex through 2100. For years after 2100, I use the Markov population process of Braun and Joines (2015) to extend the population to a stationary distribution.

The projected population of an age group can differ from the number of survivors from the previous year. I treat this difference as net migration. I measure population relative to

$\overline{N}^W$, the number of adults aged 20–64 in 2010:

$$n_{j,t} = \frac{N_{j,t}}{\overline{N}^W}, \qquad n_t^W = \sum_{j=20}^{64} n_{j,t}, \tag{1}$$

where n_t^W is the working-age population in these units.

The old-age dependency ratio is the number of adults aged 65–89 per 100 adults aged 20–64. This ratio approximately doubles between 2010 and the stationary population (Figure A.1). Labor-augmenting productivity A_t grows at a constant rate g. I normalize it so that $A_{2010} = 1$. I express quantities in 2010 dollars. For computation, I divide quantities by A_t so that this gives a balanced-growth path in the long run.

2.2 Environment

A household of age j in year t starts the year with assets $a \geq 0$ and skill $h > 0$. It chooses consumption c, saving $x \geq 0$, and hours of work $\ell \in [0, 1)$. Hours are measured as a fraction of sixteen hours per day. Households must retire at age 65, so $\ell = 0$ from that age. Households adjust labor supply only through hours of work before age 65. Following Braun and Joines (2015), households face no idiosyncratic earnings risk.[2]

Households value consumption and leisure through a constant-relative-risk-aversion function of a Cobb–Douglas composite of the two:

$$u(c, \ell) = \frac{[c(1-\ell)^v]^{1-\sigma} - 1}{1-\sigma}$$

and the estate x they leave at death:

$$\mathcal{W}(x) = \vartheta \frac{(1 + x/b_*)^{1-\sigma} - 1}{1-\sigma}.$$

The bequest term $\mathcal{W}$ represents a warm-glow motive, as in De Nardi (2004). Households receive utility from the estate they leave. The scale b_* is set in the calibration. Bequests are a luxury good in the model. Households below a wealth threshold leave no estate. Most estates come from wealthy households, as in the data. In year $t + 1$, the value of a bequest x

[2] I exclude individual earnings shocks to focus on how saving, work, and skill accumulation respond to benefit rules and their financing. This assumption also makes repeated transition and compensation calculations manageable. Assets and skills are the only household state variables. Earnings risk would give households a precautionary saving motive (Conesa and Krueger, 1999; Nishiyama and Smetters, 2007). It could also change their ability to absorb benefit cuts. Thus, the quantitative valuations depend on this assumption.

is $\mathcal{W}_{t+1}(x) = A_{t+1}^{1-\sigma}\mathcal{W}(x/A_{t+1})$. This adjustment makes the bequest motive consistent with balanced growth.

Skills increase through learning by doing, as in Imai and Keane (2004) and Chang et al. (2002):

$$h_{j+1,t+1} = (1-\delta_h)h_{j,t} + \eta\,\ell_{j,t}h_{j,t}^{\gamma} \tag{2}$$

where the parameter δ_h is the rate of skill depreciation. Work produces current earnings. It also builds skills that increase future earning capacity. The parameter η measures the efficiency of learning by doing. The parameter γ determines how the skill gain depends on existing skills. The marginal effect of work time on next year's skill is $\partial h_{j+1,t+1}/\partial \ell_{j,t} = \eta h_{j,t}^{\gamma}$. Younger workers have more years before retirement to earn a return on this additional skill.

Learning by doing links current work incentives to future earnings. A household can work more to prepare for a benefit cut, building experience as well as current income. Higher benefits and the taxes used to finance them can reduce hours, skills, and future earnings. The model captures these responses through the hours choice. It does not include a separate training decision.

For a household of age j in year t with assets a and skill h, the budget constraint is

$$\begin{aligned} \underbrace{(1+\tau_t^c)c}_{\text{consumption}} + \underbrace{x}_{\text{saving}} + \underbrace{o_{j,t}}_{\substack{\text{medical}\\ \text{bills}}} &= \underbrace{[1+(1-\tau^k)r_t](a+e_{j,t})}_{\text{assets and inheritance, after tax}} \\ &\quad + \underbrace{(1-\tau^{ss}-\tau^G)w_t h\ell}_{\text{after-tax earnings}} \\ &\quad + \underbrace{b_{j,t}+s_{j,t}}_{\substack{\text{Social Security}\\ \text{and other transfers}}} + \underbrace{f_{j,t}(a)}_{\substack{\text{floor}\\ \text{payment}}}, \end{aligned} \tag{3}$$

where $o_{j,t}$ is the out-of-pocket share p_j^O of age-specific medical spending. The term $e_{j,t}$ is the inheritance received. The wage per efficiency unit is w_t. The Social Security benefit is $b_{j,t}$.[3] Other cash transfers are $s_{j,t}$, and the consumption-floor payment is $f_{j,t}(a)$.

Assets, skill, and choices differ across households within an age group. Medical bills, inheritances, benefits, and other cash transfers are the same for everyone in an age group in a given year. Floor payments also depend on household assets. The consumption tax rate is τ_t^c. The Social Security payroll tax rate is τ_t^{ss}. The earnings tax rate is τ^G, and the capital-income tax rate is τ^k. The earnings and capital-income tax rates remain at their 2010

[3]Benefit amounts in the model are net of the federal income tax on benefits. This tax revenue is credited to the trust fund, so it cancels out of the fund's accounts. Appendix A.3 gives the rate used.

values. The consumption tax adjusts when the Treasury must balance its budget. From age 65, the government guarantees consumption of at least $\underline{c}_t = A_t \underline{c}_0$. If a retiree's resources are less than $(1+\tau_t^c)\underline{c}_t$, the retiree receives the difference as the floor payment $f_{j,t}(a)$. The retiree consumes $\underline{c}_t$ and saves nothing. While the retiree receives support, each additional dollar of resources reduces the floor payment by one dollar. This means-tested consumption guarantee follows Hubbard et al. (1995) and De Nardi et al. (2010).

Recursive Formulation The household solves

$$V_{j,t}(a,h) = \max_{c,x,\ell} \Big\{ u(c,\ell) + \beta \Big[\pi_{j,t} V_{j+1,t+1}(\kappa_{j,t} x, h_{j+1,t+1}) + (1-\pi_{j,t}) \mathcal{W}_{t+1}(x) \Big] \Big\}, \tag{4}$$

subject to (2), (3), and the consumption floor. Next year's assets are $a' = \kappa_{j,t} x$, where $\kappa_{j,t} = \pi_{j,t} n_{j,t} / n_{j+1,t+1}$ adjusts for migration. The numerator gives the number of survivors from the current age group. The denominator gives the size of that group next year, including net migrants.

Without migration, these numbers are equal. Thus, $\kappa_{j,t} = 1$, and a survivor carries all saving into the next year. With net in-migration, $\kappa_{j,t} < 1$. The model assumes that newcomers bring no assets. It pools the survivors' saving across the larger group. This assumption reduces assets per person and preserves the group's total saving.

Three features of the solution affect the results. First, the household compares the leisure cost of an additional hour of work with its wage return and skill return. An earnings tax reduces both returns to work. Second, if next year's consumption floor binds, the marginal value of saving is zero. Third, skills and work are complementary. More skill increases the return to work. Work also builds additional skill. With the consumption floor, these effects mean that the lifetime problem need not be concave.

2.3 Estates and firms

Assets left by households that die during the year are pooled. The pool is distributed the next year. This rule follows the distribution of accidental bequests in Huggett (1996) and Braun and Joines (2015). Each age group receives a share ς_j of the pool, with $\sum_j \varsigma_j = 1$. This share is divided equally among the members of that age group. The shares differ across ages and remain fixed over time. They are based on the ages at which people received inheritances

in the 2010 Survey of Consumer Finances:

$$\mathcal{E}_{t+1} = \sum_j n_{j,t}(1-\pi_{j,t}) \int x_{j,t}\, d\mu_{j,t}, \qquad e_{j,t} = \varsigma_j \mathcal{E}_t / n_{j,t}, \tag{5}$$

where $\mu_{j,t}$ is the distribution of assets a and skill h among households of age j.

Competitive firms use capital K_t and efficiency units of labor L_t to produce output: $Y_t = K_t^{\alpha}(A_t L_t)^{1-\alpha}$. Factor prices equal marginal products. The wage is $w_t = (1-\alpha)Y_t/L_t$. The net return on capital is $r_t = \alpha Y_t/K_t - \delta$, where δ is the depreciation rate.

2.4 Social Security

The model represents the combined OASDI program with one trust fund. It follows the Trustees' combined-fund projections (Social Security Trustees, 2026). The program receives revenue from a 12.4% payroll tax on earnings up to the taxable maximum. Some earnings exceed this limit, and some are not covered by the program. Thus, the model applies the effective rate $\tau_t^{ss} = 0.124\,\zeta_t$ to all earnings where the term ζ_t is the projected share of labor income subject to the tax. Payroll receipts are $\mathcal{P}_t = \tau_t^{ss} w_t L_t$. I use calligraphic letters for program aggregates and lowercase letters for amounts per person.

A retiree of age j in year t turned 65 in year $s = t - j + 65$. That cohort's scheduled initial retired-worker benefit is $\hat{b}_s$, which grows with average earnings.[4] The initial benefit remains constant in real terms for life. It depends on cohort-average taxable earnings, not on the recipient's own earnings history. An individual therefore cannot increase their own benefit by working more. Coverage rates and the benefit tax convert initial benefits into the scheduled retired-worker benefit per member of age group j, $b_{j,t}^R$. Appendix A.3 describes these adjustments and the benefits already in payment in 2010. Other scheduled benefits per member, $b_{j,t}^O$, follow their projected share of GDP and age allocation.[5] The aggregates are $\mathcal{H}_t^R = \sum_j n_{j,t} b_{j,t}^R$ and $\mathcal{H}_t^O = \sum_j n_{j,t} b_{j,t}^O$.

A multiplier m_t scales the retired-worker component. The generosity experiments choose

[4]The initial benefit is $\hat{b}_s = \varphi_s \rho\, \bar{e}_{s-1}$. Here $\bar{e}_{s-1}$ is average taxable earnings per covered worker in the year before the cohort turns 65, ρ is the ratio of the average initial benefit of newly entitled retired workers to these earnings, and φ_s captures the rise in the normal retirement age for cohorts born in 1960 or later. Because the model has no taxable maximum or covered-worker count, $\bar{e}_t$ is proportional to the model's wage bill per working-age adult, $w_t L_t / n_t^W$, so initial benefits grow one-for-one with wages per working-age adult. Appendix A.3 gives the values and their construction.

[5]Other scheduled benefits are the OASDI cash benefits not paid to retired workers: disabled-worker benefits, benefits to spouses and children of retired and disabled workers, and survivor benefits. They keep their projected share of scheduled OASDI cost and their 2010 allocation across ages (Appendix A.4). The benefit multiplier m_t below does not apply to them.

it, and all other policies set $m_t = 1$. Scheduled spending is

$$\mathcal{H}_t = m_t \mathcal{H}_t^R + \mathcal{H}_t^O. \tag{6}$$

The multiplier applies to existing retirees as well as new beneficiaries. It is not a replacement rate or a share of total retirement income.

The fund holds reserves $\mathcal{F}_t$ in Treasury securities earning gross return R_t^F. It receives payroll receipts and Treasury transfers $\mathcal{J}_t$. It pays other program expenses $\mathcal{O}_t$ before benefits. The fund cannot borrow. Thus, it pays only the fraction ϕ_t of scheduled benefits that its resources can finance:

$$\phi_t = \min\left\{1, \frac{R_t^F \mathcal{F}_t + \mathcal{P}_t + \mathcal{J}_t - \mathcal{O}_t}{\mathcal{H}_t}\right\}, \tag{7}$$

$$\mathcal{F}_{t+1} = R_t^F \mathcal{F}_t + \mathcal{P}_t + \mathcal{J}_t - \phi_t \mathcal{H}_t - \mathcal{O}_t \ \geq 0. \tag{8}$$

Each member of age group j receives benefits $b_{j,t} = \phi_t(m_t b_{j,t}^R + b_{j,t}^O)$.

The policies differ in m_t and in what $\mathcal{J}_t$ covers. Under the baseline, $m_t = 1$ and $\mathcal{J}_t$ contains only the specified reimbursements, so benefits fall when the fund can no longer pay the full schedule. I call these *funding-linked benefit cuts*. Under full benefits, $m_t = 1$ and $\phi_t = 1$ in every year: $\mathcal{J}_t$ also covers any shortfall. The generosity experiments likewise pay the selected schedule in full, so $\phi_t = 1$. In these experiments, $\mathcal{J}_t$ includes shortfall funding and any reserve sweep back to the Treasury. Other benefit rules and the retirement consumption floor are unchanged, although their outlays may respond to output and household choices.

2.5 Government spending and the Treasury

Government spending outside Social Security, Z_t, includes purchases of goods and services, cash transfers, Medicare spending $\mathcal{M}_t Y_t$, and consumption-floor payments. Purchases per resident and cash transfers per recipient grow with productivity. The Medicare share of output, $\mathcal{M}_t$, follows an exogenous path, as in Attanasio et al. (2011) and Braun and Joines (2015). Households also pay out-of-pocket medical expenses which vary by age.

The Treasury collects income-tax revenue

$$T_t = \tau^G w_t L_t + \tau^k r_t (K_t + B_t).$$

It also collects consumption-tax revenue $\tau_t^c C_t$. After Social Security is combined with the

rest of the government, debt evolves as

$$B_{t+1} = (1 + r_t)B_t + \underbrace{\phi_t \mathcal{H}_t + \mathcal{O}_t - \mathcal{P}_t}_{\text{Social Security deficit}} + \underbrace{Z_t - T_t - \tau_t^c C_t}_{\substack{\text{non-Social-Security} \\ \text{primary deficit}}}. \tag{9}$$

The Social Security deficit equals benefits plus other program costs, less payroll receipts. Treasury transfers and reserve sweeps cancel on consolidation. The primary deficit of the rest of the government equals purchases, transfers, Medicare, and consumption-floor payments, less income-tax and consumption-tax revenue.

The consumption tax limits next year's debt to $\bar{b}Y_{t+1}$. If debt would remain within the limit without this tax, the tax rate is zero. Otherwise, the tax raises the revenue needed to respect the ceiling:

$$\tau_t^c = \max\left\{0, \frac{\widetilde{B}_{t+1} - \bar{b}Y_{t+1}}{C_t}\right\}, \tag{10}$$

where $\widetilde{B}_{t+1}$ is next year's debt with a zero consumption tax. The baseline limit is $\bar{b} = 1$, or 100% of GDP. It is an upper bound, not a requirement to hold debt at the ceiling. The government can accumulate assets when revenue exceeds spending and the consumption tax is zero. In the generosity experiments, the Treasury sweeps trust-fund reserves into the consolidated account after the reform. These internal transfers cancel on consolidation. Lower benefits can therefore lower taxes, repay debt, or build government assets. Unused payroll revenue is not assumed to accumulate in a separate trust fund. This fiscal rule follows a debt target in the spirit of Bohn (1998) where the consumption tax enforces the target. Kitao (2014) studies this tax as one way to restore solvency.

In the model, the government uses a consumption tax to finance budget shortfalls when further borrowing would exceed the debt limit. These shortfalls can arise from aging, Medicare spending, or the full-benefit policy. This tax is a standard choice in models of aging and government debt (Altig et al., 2001; Braun and Joines, 2015; Hansen and İmrohoroğlu, 2016; İmrohoroğlu et al., 2016). It distorts workers' incentives to work, but retirees also pay the tax. Thus, retirees bear part of the cost of their own benefits.[6]

[6] Nishiyama and Smetters (2005) show that replacing a progressive income tax with a flat consumption tax can reduce efficiency when households cannot insure against wage shocks. That channel is absent here. The model has no individual earnings risk. It also adds the consumption tax to existing income taxes instead of replacing them.

2.6 Equilibrium

The model uses a competitive equilibrium with perfect foresight along a transition path. This concept follows Auerbach and Kotlikoff (1987), Ríos-Rull (1996), and Huggett (1996). Households and firms take prices and policy rules as given. Their expectations about future prices and aggregate quantities match the paths produced by their choices. Tax revenues depend on the resulting earnings, capital income, and consumption. The government's budget and financing rules hold in every year.

Let $t_0 = 2010$. Let $\mathbf{s} = (a, h)$ be the state of a household of age j in year t. The state consists of assets a and skill h. The household's state in the next year is

$$\mathbf{s}' = (a', h') = \mathbf{T}_{j,t}(\mathbf{s}) = \Big(\kappa_{j,t} x_{j,t}(\mathbf{s}),\ (1 - \delta_h)h + \eta \ell_{j,t}(\mathbf{s}) h^{\gamma}\Big).$$

Here $x_{j,t}(\mathbf{s})$ and $\ell_{j,t}(\mathbf{s})$ are the household's saving and hours of work, given its state. The resulting assets a' and skill h' form its state at age $j+1$ in year $t+1$.

Definition 1 (Perfect-foresight Competitive Equilibrium)**.** *Take as given preferences, technology, demographic paths $\{n_{j,t}, \pi_{j,t}\}$, productivity growth g, and the benefit, coverage, spending, Medicare, and consumption-floor rules. Also take as given income-tax rates (τ^G, τ^k), the payroll-tax path τ_t^{ss}, and the debt ceiling $\bar{b}$. Initial conditions are K_{t_0}, B_{t_0}, $\mathcal{F}_{t_0}$, retirement benefits already in payment, $\mathcal{E}_{t_0} = 0$, and distributions $\{\mu_{j,t_0}\}_{j=20}^{89}$ that satisfy $\sum_j n_{j,t_0} \int a \, d\mu_{j,t_0} = K_{t_0} + B_{t_0}$.*

A perfect-foresight competitive equilibrium is a sequence for $t \geq t_0$ of household policies and values $\{c_{j,t}, x_{j,t}, \ell_{j,t}, V_{j,t}\}$, distributions $\{\mu_{j,t}\}$, prices $\{w_t, r_t\}$, aggregates $\{K_t, L_t, Y_t, I_t, C_t, \mathcal{E}_t\}$, and fiscal outcomes $\{\tau_t^c, \phi_t, B_t, \mathcal{F}_t, Z_t\}$. The sequence satisfies the following conditions:

1. Households optimize. *The value functions $V_{j,t}$ and household policies solve (4). They satisfy (2), (3), the consumption floor, and $x \geq 0$. Hours of work satisfy $\ell_{j,t} = 0$ for $j \geq 65$. Households face the equilibrium prices, taxes, transfers, and inheritances.*

2. Distributions follow from household choices. *Apply this year's household policies to the distribution at age j to obtain next year's distribution at age $j+1$. For $20 \leq j < 89$ and any set of states $\mathcal{D}$,*

$$\mu_{j+1,t+1}(\mathcal{D}) = \int \mathbf{1}\{\mathbf{T}_{j,t}(\mathbf{s}) \in \mathcal{D}\} \, d\mu_{j,t}(\mathbf{s}).$$

Thus, the share of next year's $(j+1)$-year-olds in $\mathcal{D}$ equals the share of this year's

j-year-olds whose choices produce a next-year state in $\mathcal{D}$. Survival affects the size of the age group through $n_{j+1,t+1}$.

3. Firms optimize, and aggregates follow from household choices. *Factor prices and output satisfy $w_t = (1-\alpha)Y_t/L_t$, $r_t = \alpha Y_t/K_t - \delta$, and $Y_t = K_t^{\alpha}(A_t L_t)^{1-\alpha}$. Aggregate labor and consumption are*

$$L_t = \sum_{j=20}^{64} n_{j,t} \int h\ell_{j,t}\, d\mu_{j,t}, \quad C_t = \sum_{j=20}^{89} n_{j,t} \int c_{j,t}\, d\mu_{j,t}.$$

Estates satisfy (5).

4. Government rules hold. *The baseline fund follows* (7)–(8). *Full-payment policies instead set $\phi_t = 1$, apply the specified m_t, and include Treasury transfers and the chosen reserve rule. The Treasury satisfies* (9). *It sets the consumption tax according to* (10). *The equivalent conditions are*

$$\tau_t^c \geq 0, \qquad B_{t+1} \leq \bar{b} Y_{t+1}, \qquad \tau_t^c \big(\bar{b} Y_{t+1} - B_{t+1}\big) = 0. \tag{11}$$

5. Markets clear. *The asset market satisfies*

$$K_t + B_t = \sum_{j=20}^{89} n_{j,t} \int a\, d\mu_{j,t} + \mathcal{E}_t, \qquad K_{t+1} + B_{t+1} = \sum_{j=20}^{89} n_{j,t} \int x_{j,t}\, d\mu_{j,t}. \tag{12}$$

Investment is $I_t = K_{t+1} - (1-\delta)K_t$. The goods market satisfies

$$Y_t = C_t + I_t + G_t, \tag{13}$$

where G_t is all other spending on goods and services: government purchases, Medicare $\mathcal{M}_t Y_t$, households' out-of-pocket medical spending, and Social Security administration and rehabilitation services.

Few explanations are needed. First, Social Security reserves are claims between government accounts. Thus, they do not appear in (12). Treasury reimbursements to the fund cancel when the government accounts are consolidated. Second, a cash transfer gives the household money to spend or save. The transfer itself does not use goods or services. Consumption financed by the transfer is already included in C_t. Adding cash transfers to (13) would double-count this consumption. The goods-market equation follows from combining the household and

government budgets with asset-market clearing. Third, the equilibrium determines when Social Security reserves are exhausted, when the debt ceiling binds, and the long-run share of scheduled benefits that the program can pay.

Definition 2 (Balanced-growth equilibrium)**.** *A balanced-growth equilibrium is a perfect-foresight competitive equilibrium with stationary demographic inputs. Within each age group, the distribution of* $(a/A_t, h)$ *is constant. Hours, skills, the interest rate, tax rates, and* ϕ_t *are also constant. Output, capital, consumption, wages, benefits, and fiscal stocks grow at rate* g*.*

The transition equilibrium follows Definition 1. It starts from the 2010 initial conditions and approaches a balanced-growth equilibrium.

3 Calibration

I calibrate the model to the U.S. economy in 2010. Household parameters are chosen in a stationary *reference economy*. Mortality rates and fiscal rules remain at their 2010 settings, and quantities are detrended by productivity growth. The Social Security trust fund has no reserves in this economy.

The *baseline transition* determines the state inherited by fiscal reforms in 2026. It starts from the 2010 population, public debt, and trust-fund reserves, together with the calibrated capital stock and household distribution. Projected demographics and program rules then determine the path to 2026 and beyond. Appendix A.2 gives the sources and construction of the calibration inputs.

3.1 Externally assigned parameters

The production parameters match standard capital and return benchmarks. I set the capital share to $\alpha = 1/3$. Annual productivity growth is $g = 1.13\%$, the Trustees' assumption for real-wage growth. Capital depreciation is 6.61% per year, which gives a pre-tax return of 4.5% at a capital–output ratio of 3. Table A.1 reports the externally assigned parameters and their sources.

Household utility has curvature $\sigma = 2$. The bequest scale equals the average retired-worker benefit in payment in 2010. This scale fixes the units of the bequest motive; its strength is chosen in the joint calibration.

The skill exponent $\gamma = 0.05$ fixes the curvature of learning by doing, with Fan et al. (2024) as the benchmark (Appendix A.3). The exponent remains fixed when I fit learning efficiency

Table 1: Jointly calibrated household parameters

Parameter		Value	Principal target	Source	Moment	
					Data	Model
Discount factor	$\widetilde{\beta}$	0.9986	Capital–output ratio	Literature	3.00	3.00
Leisure weight	v	3.820	Hours per day, out of 16	ATUS	3.93	3.93
Learning coefficient	η	0.4507	Wage ratio, ages 40–44 / 20–24	CPS-ORG	1.905	1.905
Skill depreciation	δ_h	0.0471	Wage ratio, ages 60–64 / 40–44	CPS-ORG	1.007	1.007
Bequest strength	ϑ	150.3	Estates / GDP (%)	SCF, SSA	5.38	5.38

Notes: The five parameters are fitted jointly. Rows associate each parameter with its principal economic role. Literature: De Nardi et al. (1999) and Cagetti and De Nardi (2006). ATUS: mean work hours per calendar day in the 2010 American Time Use Survey, for all adults aged 20–64. The mean includes nonworkers and zero-work days. The pooled mean is the hours target. CPS-ORG: 2010 Current Population Survey outgoing rotation groups, workers in the indicated age groups. SCF, SSA: Survey of Consumer Finances wealth and Social Security mortality data. The model estate moment uses reference-economy age profiles, weighted by the 2010 population and scaled to initial private wealth. Appendix A.2 gives sample definitions and moment construction.

and skill depreciation.

The capital-income tax rate is 30%, the effective rate used by Kitao (2014).

3.2 Joint calibration to 2010 moments

Five household parameters jointly match saving, work, skill accumulation, and bequests in the reference economy. I choose the discount factor, leisure weight, learning efficiency, skill depreciation, and bequest strength to match five moments. The targets are the capital–output ratio, average work hours, two age–wage ratios, and estates relative to GDP. At each trial parameter vector, I solve household choices and the stationary distribution and impose market clearing. Table 1 reports the fitted parameters and moments.

The capital–output target links household patience to aggregate saving. The fitted primitive discount factor is $\beta = 1.0099$. Adjustment for productivity growth gives an effective factor of $\widetilde{\beta} = 0.9986$. This factor governs private choices. The social discount rate used in policy comparisons governs the weights on future entrants.

The hours target sets the overall level of labor supply. In the 2010 American Time Use Survey, adults aged 20–64 work an average of 3.93 hours per calendar day. The mean includes nonworkers and zero-work days. I match this pooled mean in the reference economy; hours at each age remain model outcomes.

The estate target measures the annual wealth left by people who die. I combine SCF wealth by age with SSA death probabilities and scale wealth to initial capital plus public debt.

The resulting flow is 5.38% of GDP. I construct its model counterpart from reference age profiles with the same 2010 population weights and wealth scaling. This moment determines the common bequest strength.

Human capital and the age–wage profile

Learning by doing generates the age–wage profile and transmits work incentives into future earnings. The model has no separate age-productivity schedule. Work raises skill under (2), while depreciation reduces it. The observed wage profile calibrates the skill technology used to evaluate fiscal policy.

The joint calibration gives learning efficiency $\eta = 0.4507$ and annual skill depreciation $\delta_h = 0.0471$. New learning is proportional to work time. Proportional skill growth is

$$\frac{h_{j+1} - h_j}{h_j} = \eta \ell_j h_j^{\gamma - 1} - \delta_h. \tag{14}$$

For a given amount of work, the percentage gain in skill falls as existing skill rises. Depreciation removes a constant fraction of skill each year. Mean skill among age-20 entrants is normalized to one.

Two wage ratios summarize the rise and flattening of wages over the working life. From mean hourly wages by five-year age group in the 2010 CPS outgoing rotation groups (Appendix A.5), the early-career target is the ratio for ages 40–44 to ages 20–24, 1.905, and the later-career target is the ratio for ages 60–64 to ages 40–44, 1.007.[7]

Model wage means use the same positive-hours restriction as the CPS sample. I average hourly wages within each five-year age group using the reference population weights. Each household's weight reflects the number of workers it represents. Hours affect wages through skill accumulation and worker selection, and do not enter the averaging weights. Taking ratios removes the common wage per unit of skill and the dollar conversion. The two targets jointly determine learning efficiency and depreciation within the full parameter fit.

The calibrated technology reproduces the rise in wages before middle age and the plateau that follows. Figure 1 compares all nine age groups after dividing each profile by its age-20–24 mean. The model ratios are 1.905 for middle-aged relative to young workers and 1.007 for older relative to middle-aged workers. The six age groups outside the targets provide an additional comparison. The model understates wage growth at ages 25–39 and slightly overstates wages at ages 45–59. The largest relative gap is 8.5% at ages 25–29.

[7]The targets are cross-sectional means. The calibration does not estimate separate cohort effects or target wage dispersion at older ages. Skill paths are deterministic conditional on initial skill and household choices.

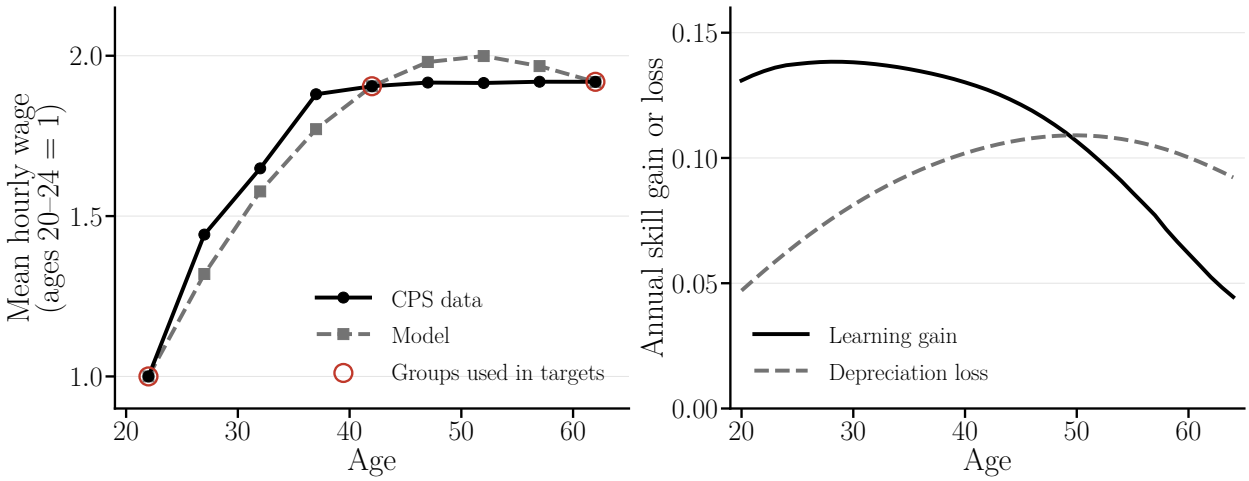


Figure 1: Human capital calibration and the age–wage profile

Notes. Left: mean hourly wages in the 2010 CPS outgoing rotation groups and the stationary reference economy. CPS means use earnings weights. Model means use stationary population weights and include households with positive work hours. Each series is divided by its own mean at ages 20–24. Points mark five-year age groups at their midpoints. Red circles mark the groups used in the two calibration targets. Right: mean skill gained through work and mean skill lost through depreciation at each age. Both means include all households of that age. Skill units set mean entrant skill to one. The difference between learning and depreciation is the change in mean skill.

The balance between learning and depreciation explains the wage profile. Learning exceeds depreciation early in life. As skill accumulates and hours fall, the learning gain declines relative to depreciation. The two components cross near age 50, when mean skill peaks. Depreciation then exceeds learning. Worker selection moderates the wage decline: households that stop working leave the wage sample but remain in the mean skill calculation.

3.3 Initial conditions in 2010

The initial population determines the demographic pressure along the baseline transition. I measure population in units of 187 million, the number of adults aged 20–64 in 2010. Population and survival probabilities follow the Trustees' projections. The old-age dependency ratio rises from 20.7 in 2010 to 45.3 in the terminal population reached in 2190.

Medical spending follows age-specific expenditure data. I construct Medicare spending and out-of-pocket medical costs from the National Health Expenditure Accounts. Future Medicare spending follows the 2025 Medicare Trustees projection.

Initial capital is three times GDP, while public debt and trust-fund reserves take their observed 2010 values. The undistributed estate pool starts at zero. Table 2 reports these stocks and their 2026 counterparts.

Household heterogeneity

Households share common preferences and differ in their resources. The discount factor, leisure weight, and bequest strength are the same for all households. New cohorts enter at age 20 with zero assets and different initial human capital. Initial skill differences produce differences in wages, work hours, and saving. Wealth and skill then evolve jointly through household choices.

Households older than 20 at the start of 2010 also differ in initial wealth. I set this opening wealth dispersion from the SCF and initial skill dispersion from CPS wages. Mean assets and skills by age follow the reference economy. These initial wealth differences supplement the skill differences that generate heterogeneity among new entrants. Figure 2 shows the resulting initial distribution.

The SCF determines the shape of the asset distribution within each age group (Appendix A.5 describes the sample). For each five-year age group, I match the zero-asset share and the standard deviation of log positive wealth with a mass at zero and a lognormal positive part. Zero-asset shares are 30.0% at ages 20–24 and 5.1% at ages 65–69. Log standard deviations range from 1.67 to 2.02.

Aggregate private wealth sets the level of initial assets. I apply one common scale factor to the reference economy's mean asset profile. With the actual 2010 population weights, total assets then equal capital plus public debt, or 3.530 times GDP. This scaling preserves the age profile and within-age relative dispersion. Figure 2 compares the resulting age means with SCF wealth; those means are not separate calibration targets.

Initial human capital follows a lognormal distribution around each reference age mean. I set the log standard deviation to 0.493, the CPS wage dispersion among workers aged 20–24. I apply this dispersion to all opening cohorts. Assets and skills are drawn independently within age; Appendix A.5 gives the numerical construction.

Zero-asset entry changes the youngest age group's wealth distribution. It raises the model's zero-asset share at ages 20–24 to 44.4%, compared with 30.0% in the SCF. The model reproduces the observed zero shares in the other age groups.

Inheritance links subsequent wealth accumulation to the estate pool. Estates are distributed equally within each recipient age using fixed SCF recipient-age shares. These shares determine who receives the pool. The estate-flow target in Table 1 determines the common bequest strength.

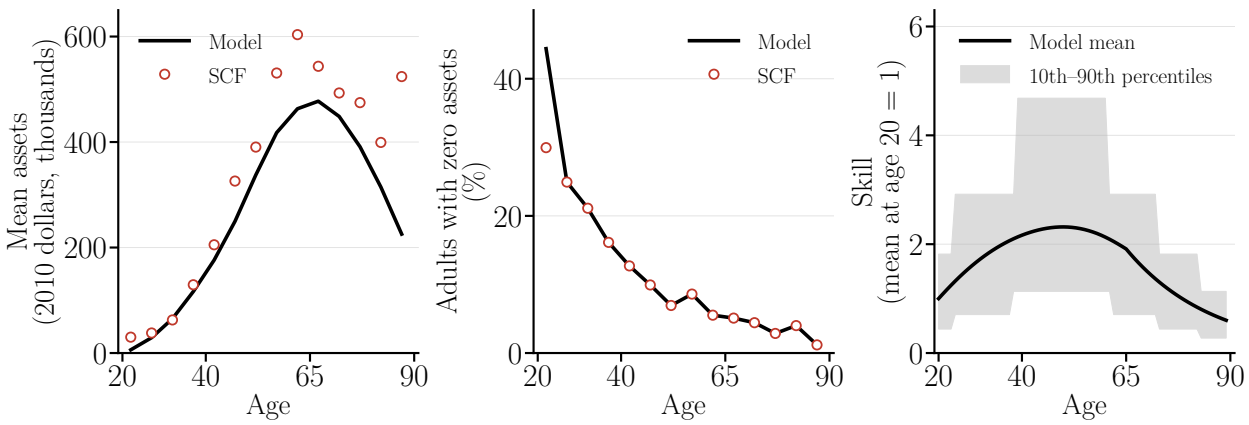


Figure 2: Household heterogeneity at the start of 2010

Notes. Left: mean nonnegative wealth per adult in the 2010 SCF and mean model assets. Center: the share of adults with zero assets. Points mark five-year age groups at their midpoints. Model means use the 2010 population weights. SCF family net worth is divided equally between spouses, and non-positive net worth is set to zero. Right: mean skill and the 10th–90th percentile range in the initial model distribution. Skill dispersion uses young-worker CPS wages. The band describes the assumed cross-sectional skill distribution; it is not a confidence interval or an individual life-cycle path. Quantiles use the inverse discrete CDF on the model grid.

3.4 Fiscal calibration

The fiscal calibration maps the 2010 accounts and the Trustees' projections into benefit and financing rules. The 2010 accounts set initial benefit levels, the payroll base, government spending, and tax rates. Projected payroll coverage and program composition determine the external fiscal paths. Figure 3 compares the baseline with the Trustees' projections. Table A.1 and Appendix A.3 report the sources.

The benefit coefficient matches newly awarded retirement benefits to taxable earnings: the annualized average benefit of newly entitled age-65 retirees relative to the preceding year's average taxable earnings per covered worker, 0.4826. A fixed scale factor maps taxable earnings per covered worker into the model's wage bill per working-age adult (Appendix A.3).

Cohort benefit schedules combine initial promises with the rules for new awards. Benefits already in payment follow the observed age profile of beneficiaries and benefits. New awards track average earnings and the statutory adjustment for claiming at age 65. Awarded benefits remain constant in real terms.

The payroll base accounts for uncovered work and earnings above the taxable maximum. Net contributions at the 12.4% statutory rate imply an initial taxable-payroll share of 0.512 (Appendix A.3).

Government purchases and transfers outside Social Security match the 2010 national

accounts. I remove Medicare from cash transfers and provide it in kind, so each expense enters the budget once.

The earnings-tax rate balances the reference government budget at its debt-to-GDP ratio. I add the required primary surplus to spending, subtract capital-income tax revenue, and divide the balance by labor income. The primary surplus covers interest payments less the new borrowing allowed by productivity growth. This gives an earnings-tax rate of 24.6%. Along the transition, the consumption tax raises the revenue needed to enforce the debt ceiling.

The consumption floor is 9.7% of 2010 GDP per adult aged 20–89 and grows with productivity. Its benchmark is the federal SSI benefit relative to the average retired-worker benefit.

Projected payroll coverage and benefit composition govern future program revenue and cost. The statutory payroll rate remains 12.4%. Effective coverage follows the Trustees' projected payroll receipts relative to labor compensation (Social Security Trustees, 2026). Retired-worker coverage and the disability, survivor, and auxiliary benefit shares follow the decomposition in Appendix A.4. This decomposition combines 2025 report shares with 2026 total-cost anchors. These inputs remain at their final values after 2100.

The fiscal targets describe the program's funding pressure. They are the reserve-exhaustion year and scheduled cost and non-interest income relative to GDP in 2100. The payable share of scheduled benefits follows from the fund budget.

The baseline reproduces the rise in Social Security cost and reserve exhaustion in 2034. Scheduled cost reaches 6.65% of GDP in 2100, compared with 6.69% in the Trustees' projection.[8] Payroll coverage and benefit composition enter as imposed projections, so this comparison measures the model's fit conditional on those inputs.

The non-interest income gap comes mainly from the tax on benefits. The model applies a fixed aggregate effective rate of 2.5%. The SSA accounting targets imply an effective rate of about 5.3% in 2100. Model income is therefore 4.29% of GDP, compared with 4.49% in the projection. The payroll normalization accounts for a smaller part of the gap.

Lower revenue reduces the share of scheduled benefits that the fund can pay. The model pays 63.0% in 2100, compared with 65.0% in the projection. Benefit cuts also reduce benefit-tax receipts. Non-interest income based on benefits actually paid is 4.23% of GDP.

[8]The exhaustion year matches the target. The within-year reserve-balance target is not matched.

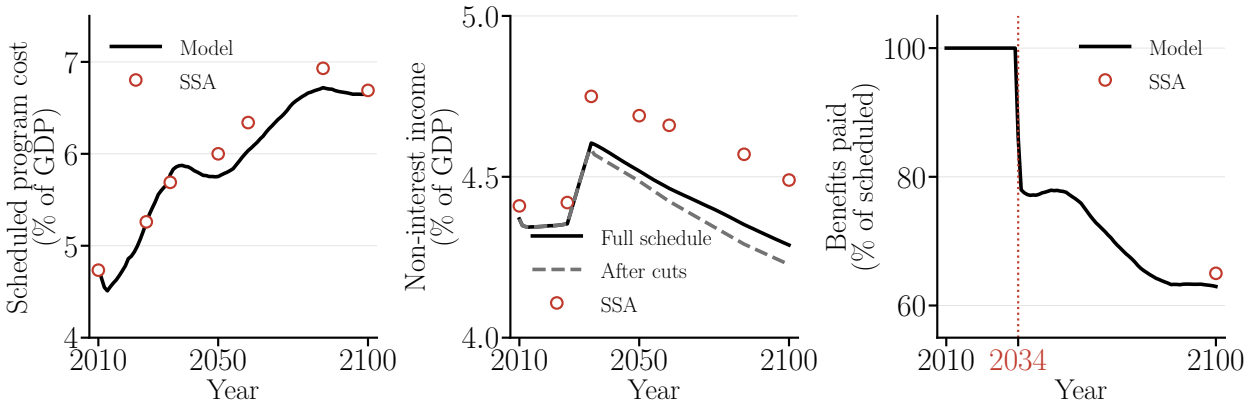


Figure 3: Fiscal calibration: accounts, projections, and model outcomes

Notes. Black lines show the baseline transition. Red circles show the 2010 accounts and the intermediate projections from the 2026 Trustees Report. Scheduled cost includes gross benefits, administration, and the railroad interchange. Non-interest income includes payroll receipts, taxes on benefits, and specified Treasury reimbursements. The black income line assumes full scheduled benefits. The dashed gray line uses benefits actually paid. The last panel shows benefits paid as a share of scheduled benefits. Its red circle marks the projected payable share in 2100. The vertical line marks reserve exhaustion in 2034; the annual model permits a partial reserve drawdown in that year. Payroll coverage and benefit composition enter as projected inputs.

3.5 The inherited reform state in 2026

All reforms within a model specification inherit the same household distribution and aggregate stocks in 2026. This state includes assets and skills by household, capital, the estate pool, public debt, and trust-fund reserves. Table 2 reports the aggregate stocks. Common initial conditions isolate the effects of the benefit rules and their financing.

Baseline expectations determine household choices before the reform announcement. Households anticipate scheduled Social Security benefits while program funding permits and funding-linked cuts after reserve exhaustion. The Treasury provides no benefit guarantee. The payroll tax remains 12.4%, and the earnings- and capital-income tax rules remain fixed. Households anticipate the consumption taxes required to enforce the 100% debt ceiling.

Reforms are announced unexpectedly in 2026. Households then have perfect foresight about the new policy and adjust consumption, saving, and work. Preferences and technology remain fixed across fiscal policies within each model specification.

Table 2: Initial stocks and the inherited reform state

	2010	2026
Capital / GDP	3.00	3.07
Public debt / GDP (%)	52.98	66.44
Trust-fund reserves / GDP (%)	17.23	8.74
Estate pool / GDP (%)	0.00	6.18

Notes: Stocks are measured at the beginning of the year. Initial capital is set to the calibration target. Initial public debt and trust-fund reserves are observed, and the estate pool starts at zero. Mean assets and skills by age follow the reference economy. Assets are scaled so that total household wealth equals capital plus public debt. The 2026 column reports the baseline state inherited by all reforms.

4 Reforming Social Security without Cohort Losses

4.1 Policy Design and Welfare Measurement

Table 3: Payroll-tax and benefit rules in the three scenarios

Policy	Payroll-tax rule	Benefit payments and financing
Baseline	12.4% in every year.	Pay full scheduled benefits through 2040, using Treasury borrowing after reserves run out in 2034. From 2041, limit benefits to program income: 60.02% of scheduled benefits in 2200.
Main reform	12.4% through 2060; then gradually declines to 3.50% in 2200.	Pay full scheduled benefits through 2040. From 2041, pay partial benefits, with further cuts as the payroll tax falls: 16.06% of scheduled benefits in 2200. Treasury-financed payroll-tax relief and benefit supplements protect transition cohorts (Table 5).
Phase-out	Gradually declines from 12.4% in 2026 to zero in 2200.	After reserves run out in 2033, pay benefits only from program income; the Treasury does not borrow to cover the program shortfall. Benefits fall to zero in 2200.

Notes. Scheduled benefits are the full pension promises before funding-related cuts. All policies are announced in 2026 and face a debt ceiling of 150% of GDP. The ceiling first binds in 2041 under the baseline and main reform, and in 2047 under phase-out. Consumption taxes finance the remaining government budget gap. Payroll-tax rates shown here exclude cohort-specific relief.

4.2 Welfare across Generations

Table 4: Main reform's headline welfare results

Measure	Main reform
Evaluated birth cohorts	804
Losing cohorts	0
Smallest cohort CEV (%)	+0.00008
Birth year of the smallest gain	1937
Cohort born 2060: CEV (%)	+0.95
Cohort born 2100: CEV (%)	+3.42
Long-run cohort CEV (%)	+7.73
Social welfare change ΔW, $\beta = 0$	+0.04
Social welfare change ΔW, $\beta = 0.97$	+0.59
Social welfare change ΔW, $\beta = 1$	+16.59
Peak public debt (% of GDP)	150

Notes. All entries compare the main reform with the debt-financed baseline at the 150% ceiling. CEVs are cohort consumption equivalents. The minimum is shown to five decimal places to avoid rounding it to zero. ΔW is measured in utility units per adult alive in 2026, with the saved stationary continuation.

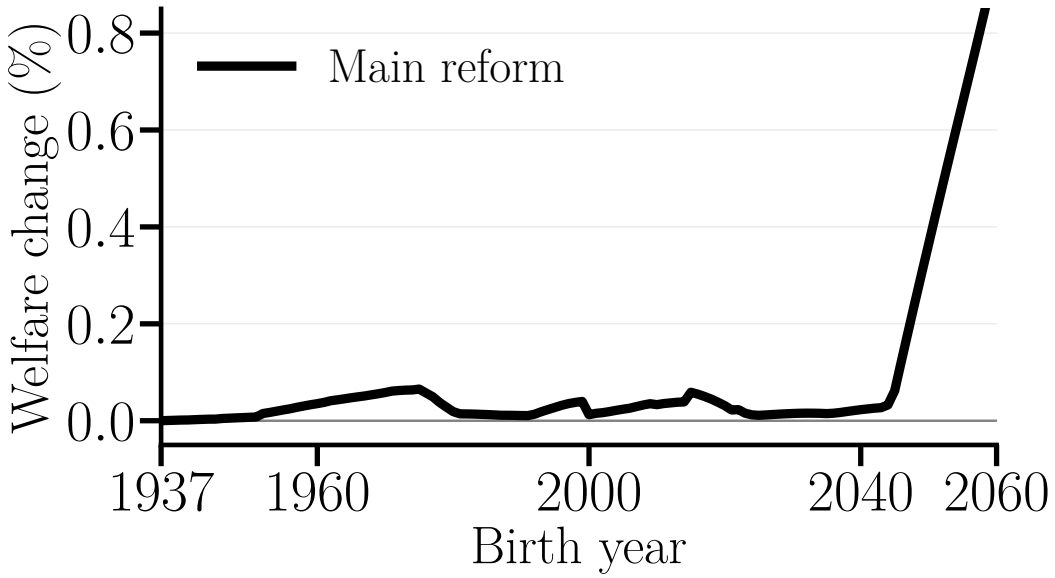


Figure 4: Welfare by birth year under the main reform

Notes. Main reform relative to the debt-financed baseline. The plot shows birth years 1937–2060 to make the transition cohorts' small gains visible. Table 4 uses all 804 evaluated cohorts. The horizontal line marks zero.

4.3 Financing the Compensation

Table 5: Compensation eligibility, timing, amounts, and financing

Instrument	Birth cohorts	Timing	Largest adjustment
Payroll-tax relief	2001–2042	From 2065 until retirement	2.29 percentage points
Benefit supplement	1981–2040	Before birth year 2000: from retirement. Later births: from 2090 or age 75, whichever comes first	8.63% of the paid benefit

Financing measure	Value
Financing source	Treasury general revenue (consumption tax)
Peak Treasury transfer	0.46% of GDP in 2065
Present value of Treasury transfers	2.15% of 2026 GDP
Maximum consumption-tax increase	0.45 percentage points in 2065
Consumption tax below baseline	From 2072 onward
Additional borrowing allowance	None; the ceiling remains 150% of GDP

Notes. The payroll-relief range lists cohorts with realized peak relief of at least 0.05 percentage points. Supplements are proportional to the benefit otherwise paid. Financing statistics cover Treasury transfers from 2041; their present value uses the saved interest-rate and output paths. Tax differences are relative to the baseline.

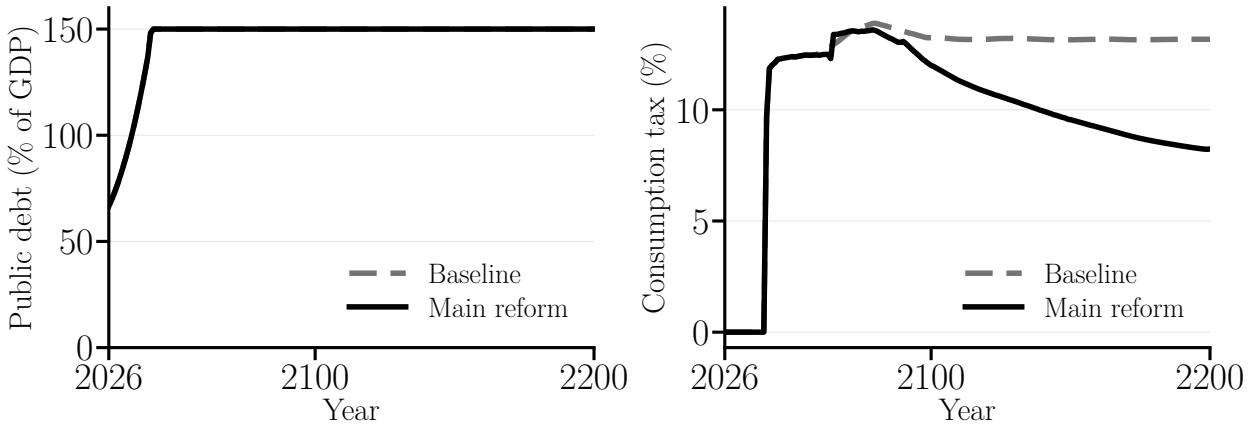


Figure 5: Public debt and consumption-tax paths

Notes. Baseline and main reform, 2026–2200. Public debt is start-of-year debt divided by each policy's GDP. Both policies face the same 150% ceiling. Consumption-tax rates are percentages of consumption expenditure.

4.4 Macroeconomic Adjustment

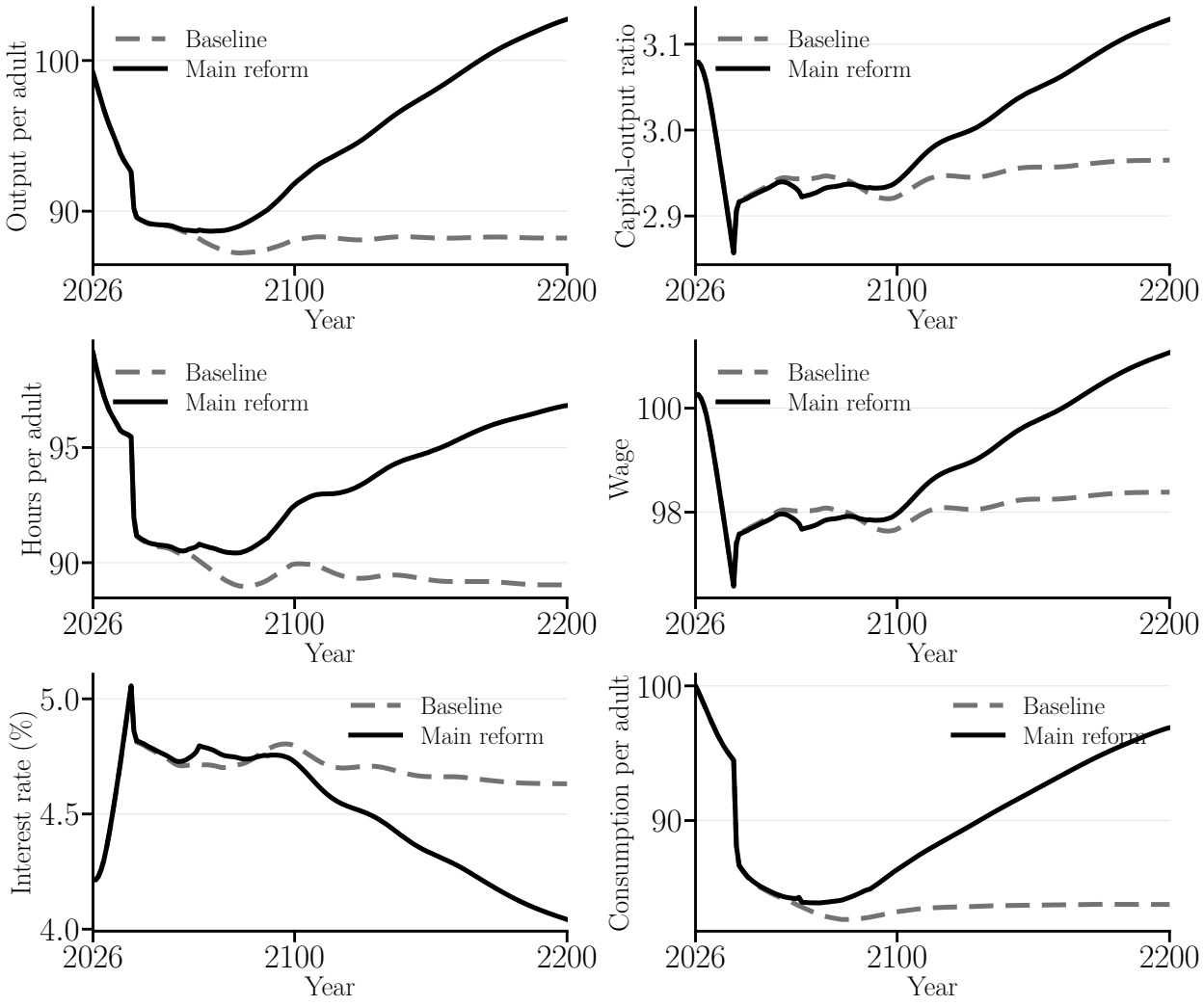


Figure 6: Macroeconomic adjustment under the main reform

Notes. Baseline and main reform, 2026–2200. Output, hours, consumption and the wage are indices with baseline 2025 = 100; output, hours and consumption are per adult. Per adult divides by the model population aged 20–89; the model has no children, so these are the model's per-capita quantities. The capital-output ratio is K/Y, and the interest rate is annual, in percent. Plotted values retain the saved paths' productivity normalization.

5 Retaining Social Security versus Phasing It Out

5.1 The Phase-Out Experiment

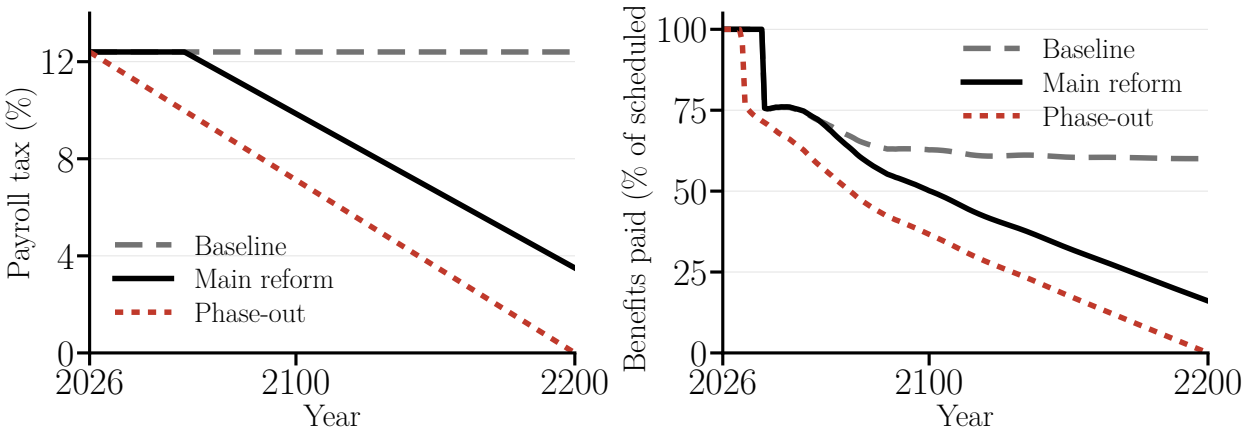


Figure 7: Payroll-tax and benefit paths across the three scenarios

Notes. The main reform's payroll line is the announced economy-wide schedule, before cohort relief. The other payroll lines use their saved effective rates. Benefits are expressed as percentages of scheduled benefits. All three scenarios face a 150% debt ceiling.

5.2 Transition Losses and Long-Run Gains

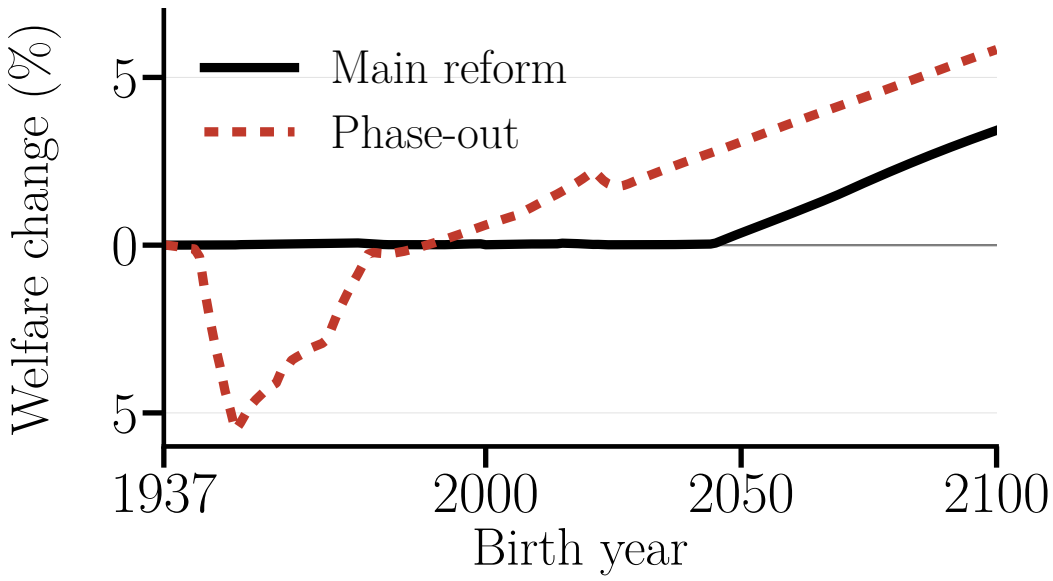


Figure 8: Cohort welfare under the main reform and phase-out

Notes. Cohort consumption equivalents relative to the common debt-financed baseline. Birth years 1937–2100 are displayed; the losing-cohort counts in Table 6 use all 804 evaluated cohorts. The horizontal line marks zero.

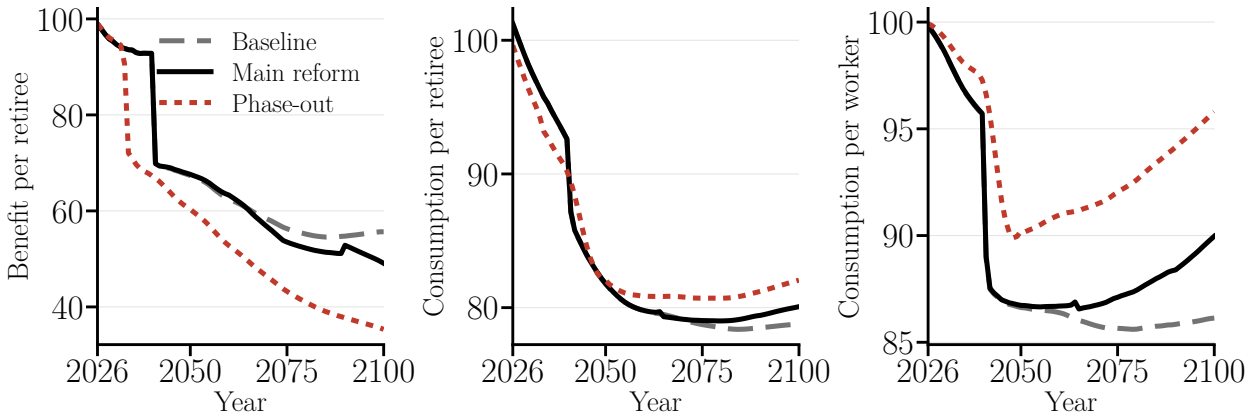


Figure 9: Benefits and consumption of retirees and workers across the three scenarios

Notes. 2026–2100; all three panels are indices with baseline 2025 = 100. Retirees are adults aged 65–89 and workers adults aged 20–64; the model population is adults aged 20–89, so per-adult quantities are the model's per-capita quantities. Benefit per retiree is Social Security outlays divided by the number of retirees. The phase-out's higher aggregate consumption is workers' consumption, because their payroll tax falls from 2026. Retirees' benefits are cut in 2033 under the phase-out, when the trust fund is exhausted without borrowing, eight years before the baseline's cut in 2041, and then fall with the payroll rate. Retirees run down assets, so their consumption falls by less than their benefits and the remaining loss falls on bequests. The retirees after 2050 are later cohorts, who saved more while working under a lower payroll tax; the lifetime losses of the cohorts alive in 2026 are in Figure 8.

Table 6: Losing cohorts, transition losses, and long-run gains

Measure	Main reform	Phase-out
Losing cohorts (of 804)	0	51
Smallest cohort CEV (%)	+0.00008	−5.52
Birth year of the smallest CEV	1937	1951
Cohort born 2060: CEV (%)	+0.95	+3.65
Cohort born 2100: CEV (%)	+3.42	+5.83
Long-run cohort CEV (%)	+7.73	+10.52

Notes. Both policies are scored against the same debt-financed baseline. CEVs are percentages of lifetime consumption, or remaining-lifetime consumption for adults alive in 2026. Counts use all 804 evaluated birth cohorts. The main reform's minimum is shown to five decimal places.

5.3 The Role of Social Welfare Weights

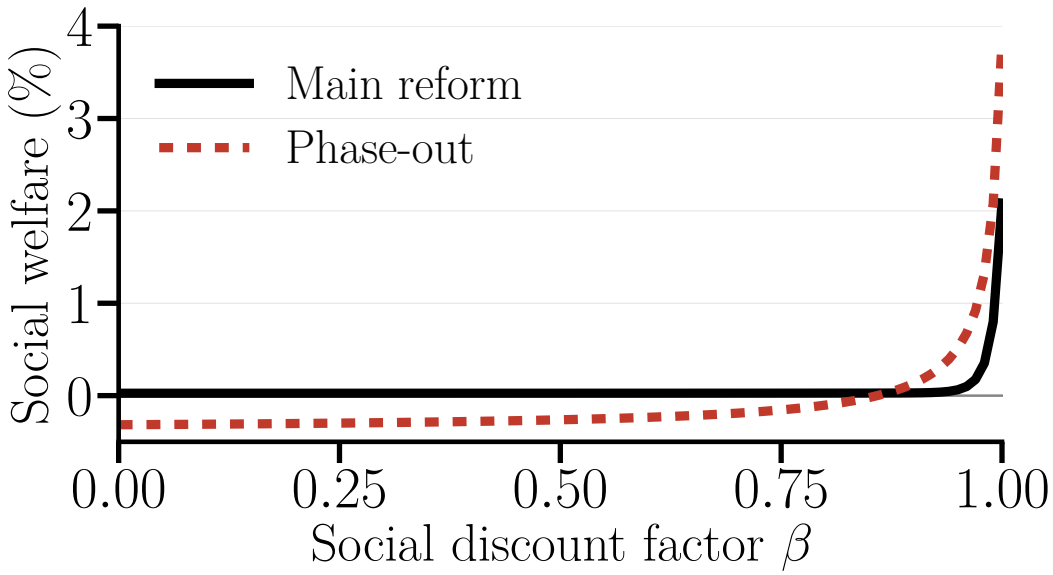


Figure 10: Social consumption equivalents across social discount factors

Notes. The plotted measure is the uniform social consumption equivalent of each policy relative to the debt-financed baseline, in percent. It uses the same population weights, growth adjustment, and stationary continuation as the utility changes in Table 7.

Table 7: Social welfare at selected discount factors

β	0.00	0.20	0.40	0.60	0.80	0.90	0.95	0.97	0.98	0.99	1.00
Main reform	+0.04	+0.04	+0.05	+0.05	+0.05	+0.07	+0.18	+0.59	+1.38	+3.99	+16.59
Phase-out	−0.54	−0.52	−0.49	−0.43	−0.20	+0.30	+1.44	+3.09	+5.14	+10.26	+29.94

Notes. Changes in utility per 2026 adult relative to the debt-financed baseline. The reference discount factor is $\beta = 0.97$. Future-cohort weights are proportional to $[\beta(1+g)^{1-\sigma}]^{t-2026}$, with $g = 1.13\%$ and $\sigma = 2$, including the saved stationary continuation after 2510.

5.4 The Long-Run Payroll Rate

Table 8: The same transition at other long-run payroll rates

Long-run payroll rate (%)	Losing cohorts	Smallest CEV (%)	Born 2060 (%)	Born 2100 (%)	Long run (%)	ΔW, $\beta = 0$	ΔW, $\beta = 0.9$	ΔW, $\beta = 0.97$	ΔW, $\beta = 1$
0	0	+0.00015	+1.46	+4.62	+10.52	+0.05	+0.07	+0.80	+22.27
2	0	+0.00011	+1.17	+3.94	+8.94	+0.05	+0.06	+0.68	+19.07
3.5 (main reform)	0	+0.00008	+0.95	+3.42	+7.73	+0.04	+0.07	+0.59	+16.59
4.8	0	+0.00007	+0.79	+2.96	+6.68	+0.03	+0.06	+0.51	+14.42
6.5	0	+0.00001	+0.46	+2.30	+5.26	+0.04	+0.06	+0.39	+11.28
8	1	−0.00	+0.22	+1.71	+3.99	+0.03	+0.04	+0.27	+8.45

Notes. Each row keeps the main reform's dates, financing rules, debt ceiling, and compensation instruments and changes only the payroll rate the schedule reaches in 2200; the relief and supplement tables are re-solved at each rate until no cohort loses. At 6.5% and 8% the compensation bands are widened to births 1937–2060 to reach the losing cohorts; at 8% the cohort born in 1937 remains at -8×10^{-7}, within numerical resolution. All rows are scored against the same debt-financed baseline; ΔW is in utility units per 2026 adult.

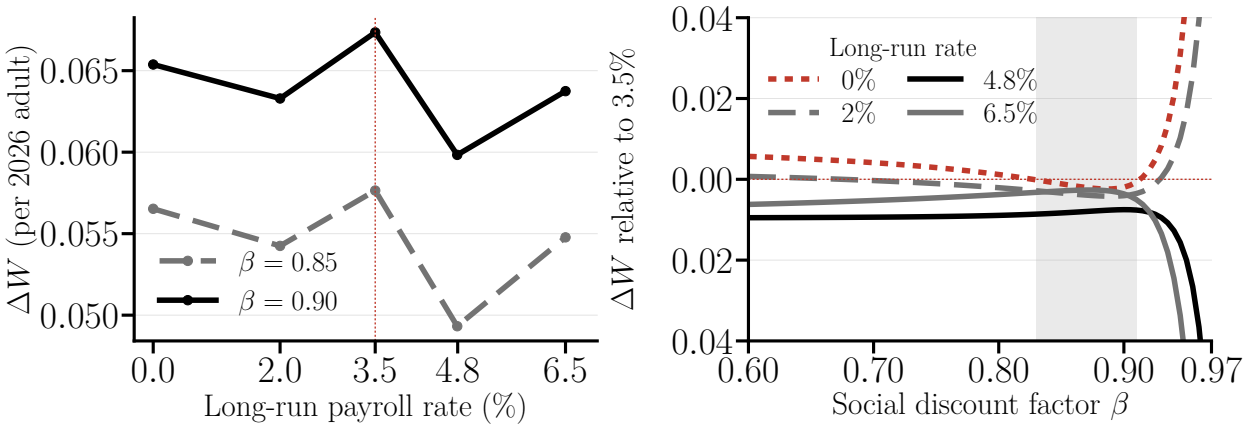


Figure 11: Social welfare across long-run payroll rates

Notes. Left: ΔW by long-run payroll rate at social discount factors of 0.85 and 0.90; the dotted line marks the main reform's 3.5%. Right: ΔW at each rate minus ΔW at 3.5%, as a function of the social discount factor; the shaded band, β from 0.83 to 0.91, is the range in which 3.5% gives the highest social welfare among the rates solved. Differences inside the band are a few thousandths of a utility unit; outside it the zero rate is highest.

Table 9: Who gains from a larger or smaller program: ΔW by cohort group and long-run payroll rate

	0	2	3.5	4.8	6.5	8
Social discount factor $\beta = 0.9$						
Alive in 2026, age 62 and over	+0.007	+0.006	+0.004	+0.003	+0.001	+0.001
Alive in 2026, under 62	+0.045	+0.041	+0.040	+0.031	+0.035	+0.027
Born 2007–2035	+0.005	+0.011	+0.019	+0.022	+0.026	+0.016
Born 2036–2100	+0.007	+0.006	+0.004	+0.003	+0.002	+0.001
Born after 2100	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000
Total	+0.065	+0.063	+0.067	+0.060	+0.064	+0.045
Social discount factor $\beta = 0.97$						
Alive in 2026, age 62 and over	+0.007	+0.006	+0.004	+0.003	+0.001	+0.001
Alive in 2026, under 62	+0.045	+0.041	+0.040	+0.031	+0.035	+0.027
Born 2007–2035	+0.012	+0.023	+0.035	+0.044	+0.045	+0.031
Born 2036–2100	+0.534	+0.434	+0.355	+0.299	+0.200	+0.129
Born after 2100	+0.206	+0.178	+0.155	+0.135	+0.107	+0.081
Total	+0.805	+0.681	+0.590	+0.514	+0.388	+0.269

Notes. Each entry is the group's contribution to ΔW in utility units per 2026 adult: the population-weighted sum of its cohorts' lifetime-utility changes, with cohorts entering after 2026 discounted by $[\beta(1+g)^{1-\sigma}]^{t-2026}$. Columns are long-run payroll rates in percent; 3.5 is the main reform. Rows sum to the total.

The table explains the shape of Figure 11. Two groups prefer the smallest program. Cohorts born after 2035 gain more the lower the rate, because they work and retire entirely under the smaller program, and at a discount factor of 0.97 their contribution dominates the sum. Adults alive in 2026 also prefer the smallest program: the cohorts that need compensation are held at zero whatever the rate, and the younger workers who need none gain more from a deeper cut. The cohorts born between 2007 and 2035 prefer a larger program. At a low long-run rate nearly all of them would lose and are compensated back to exactly zero; at a higher rate fewer of them lose, and those who do not keep a positive gain. The interior optimum at 3.5% for discount factors between 0.83 and 0.91 is the point at which this middle group's preference for a larger program offsets the two outer groups' preference for a smaller one. It is a property of the transition and of the no-losers constraint, not a long-run benefit of the program: in this economy, with deterministic earnings and health and an estate pool that leaves little longevity risk to insure, the long-run welfare of future cohorts rises monotonically as the program shrinks, as the Aaron condition predicts when the interest rate exceeds the growth rate. Positive optimal sizes in the literature come from the insurance that Social Security provides against longevity, earnings, and health risk (İmrohoroğlu et al., 1995; Conesa and Krueger, 1999; Huggett and Ventura, 1999; Hosseini and Shourideh, 2019); that channel is absent here and would add to the long-run size, while leaving the transition mechanism identified in this section unchanged.

6 Robustness of the Reform

6.1 Bequests and Retirement Resources

Table 10: Results with and without a bequest motive

	Bequest motive		No bequest motive	
	Main	Phase-out	Main	Phase-out
Losing cohorts (of 804)	0	51	0	50
ΔW, $\beta = 0$	+0.04	−0.54	+0.03	−0.36
ΔW, $\beta = 0.97$	+0.59	+3.09	+0.62	+3.85

Notes. Each economy is compared with its own debt-financed baseline. The no-bequest economy pools deceased households' assets within their birth cohort, recalibrates household parameters, and retains the retirement consumption floor. Its supplement may start in 2026 for cohorts born before 1952. ΔW is in utility units per 2026 adult.

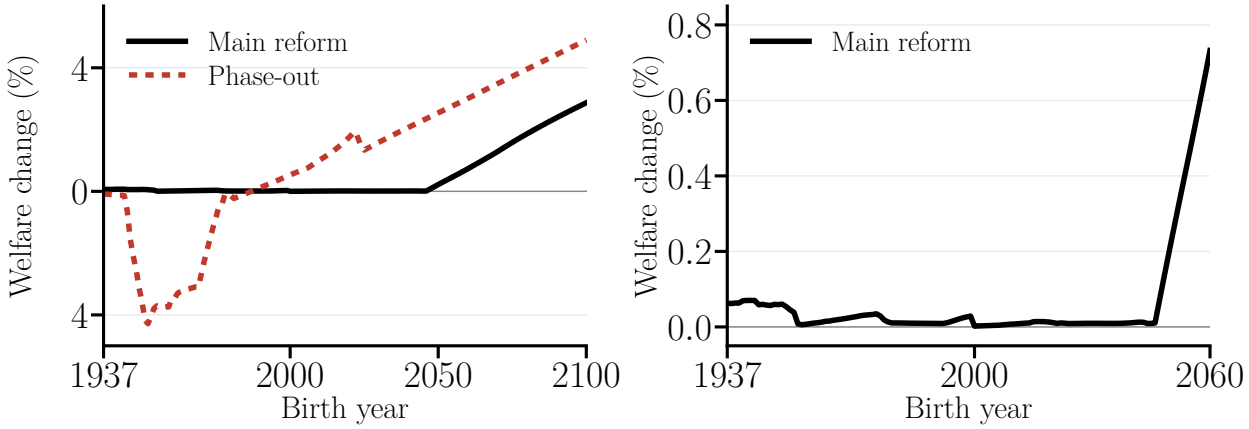


Figure 12: Cohort welfare in the recalibrated economy without bequests

Notes. Recalibrated no-bequest economy with within-cohort asset pooling and the retirement consumption floor. Each policy is compared with that economy's baseline. Left: main reform and phase-out, birth years 1937–2100. Right: main reform, birth years 1937–2060. All cohort measures are consumption-equivalent percentages.

6.2 Compensation Financing and Implementation Speed

Table 11: Alternative compensation financing and adjustment speeds

Policy variant	Losing cohorts	ΔW, $\beta = 0$	ΔW, $\beta = 0.97$
Main reform	0	+0.04	+0.59
Relief financed by a surcharge on cohorts born 2044 onward	0	+0.04	+0.55
Slower main-reform adjustment, completed in 2300	0	+0.03	+0.34
Phase-out completed in 2200	51	−0.54	+3.09
Phase-out completed in 2300	42	−0.33	+2.19
Phase-out in 2200 with the baseline tax rule	51	−0.55	+3.04

Notes. All rows use the bequest economy and its common debt-financed baseline. Counts cover 804 cohorts. ΔW is in utility units per 2026 adult. The slower-adjustment row is verified at the 1e-4 market tolerance used for robustness runs; the other rows at 5e-5. These are the saved financing and timing variants; no new policy solve is added.

6.3 Borrowing Allowances

Table 12: Compensation financed under alternative debt ceilings

Borrowing allowance	Losing cohorts	ΔW, $\beta = 0$	ΔW, $\beta = 0.97$
150% of GDP (main reform)	0	+0.04	+0.59
152% of GDP while compensation is paid	0	+0.07	+0.61
160% of GDP while compensation is paid	0	+0.12	+0.66

Notes. The higher-ceiling variants let public debt rise above 150% of GDP during the decades in which the relief and the supplement are paid and return it to 150% afterwards; the instruments are otherwise those of the main reform. Each row is scored against the same debt-financed baseline. Counts cover 804 cohorts; ΔW is in utility units per 2026 adult.

7 Conclusion

The trust fund is projected to run out in 2034. After that, every way of handling Social Security assigns a loss to some group of people, and the question is whether the loss can be avoided altogether. This paper asks whether the system can be made smaller without making any birth cohort worse off, and whether a smaller system is preferable to none.

It can be. Against a baseline in which the Treasury borrows until public debt reaches 150% of GDP and benefits then fall to what payroll revenue covers, the payroll tax can be cut from 12.4% to 3.5% with no cohort losing. The cohorts who would otherwise lose are held at zero by a lower payroll rate while they work and a higher benefit late in retirement,

both paid from general revenue and timed so that the cost falls on cohorts who gain from the cut. None of the 804 cohorts evaluated ends below its baseline welfare, and cohorts born after 2044 gain up to 7.7% of lifetime consumption.

A complete phase-out raises output and consumption by more, in the long run and already during the transition, because the extra consumption goes to workers whose payroll tax falls from 2026 and to the cohorts who follow them. Retirees' benefits are cut in 2033, eight years earlier than in the baseline, and the cohorts alive in 2026 bear that cut without the later gains. The cohort born in 1951 lose by 5.5% of remaining lifetime consumption. Counting only adults alive in 2026, retaining the system is a gain and the phase-out a loss; at a social discount factor of 0.97 the phase-out's gain is five times larger. Retention is preferred for discount factors up to 0.87, and the ranking survives when the bequest motive is removed and the economy recalibrated.

The implication is that the choice facing Congress is wider than paying full benefits or cutting them. The efficiency gains from a smaller pay-as-you-go system are large enough to compensate everyone who would lose from the transition, provided the compensation is assigned by birth cohort, financed from general revenue, and paid when the gaining cohorts make up most of the tax base. What remains after that is a distributional question, how much weight to give cohorts not yet born, and the paper's numbers locate the threshold at which that weight reverses the answer. A reform debate framed as scheduled benefits against payable benefits leaves this part of the design space unused. The long-run size of the program follows the same logic. In this economy the only reason to land on a positive payroll rate is the transition: holding every cohort at or above the baseline is cheapest, for social discount factors between 0.83 and 0.91, at a rate near 3.5%, because a shallower cut leaves more of the cohorts born between 2007 and 2035 with gains of their own instead of compensation. The insurance that Social Security provides against longevity, earnings, and health risk, the source of positive optimal sizes in the literature, is absent here and would add to that level. The paper thus isolates the transition component of the optimal size; the insurance component is the natural next step.

References

Aaron, Henry J., "The Social Insurance Paradox," *Canadian Journal of Economics and Political Science*, 1966, *32* (3), 371–374.

Altig, David, Alan J. Auerbach, Laurence J. Kotlikoff, Kent A. Smetters, and Jan Walliser, "Simulating Fundamental Tax Reform in the United States," *American Economic Review*, 2001, *91* (3), 574–595.

Attanasio, Orazio, Sagiri Kitao, and Giovanni L. Violante, "Financing Medicare: A General Equilibrium Analysis," in John B. Shoven, ed., *Demography and the Economy*, Chicago: University of Chicago Press, 2011, pp. 333–366.

Auclert, Adrien, Hannes Malmberg, Frédéric Martenet, and Matthew Rognlie, "Demographics, Wealth, and Global Imbalances in the Twenty-First Century," Working Paper 29161, National Bureau of Economic Research 2021.

Auerbach, Alan J. and Laurence J. Kotlikoff, *Dynamic Fiscal Policy*, Cambridge: Cambridge University Press, 1987.

Blanchard, Olivier, "Public Debt and Low Interest Rates," *American Economic Review*, 2019, *109* (4), 1197–1229.

Blandin, Adam, "Human Capital and the Social Security Tax Cap," *International Economic Review*, 2021, *62* (4), 1599–1626.

Bohn, Henning, "The Behavior of U.S. Public Debt and Deficits," *Quarterly Journal of Economics*, 1998, *113* (3), 949–963.

Bouton, Laurent, Alessandro Lizzeri, and Nicola Persico, "The Political Economy of Debt and Entitlements," *Review of Economic Studies*, 2020, *87* (6), 2568–2599.

Braun, R. Anton and Douglas H. Joines, "The Implications of a Graying Japan for Government Policy," *Journal of Economic Dynamics and Control*, 2015, *57*, 1–23.

Breyer, Friedrich, "On the Intergenerational Pareto Efficiency of Pay-as-you-go Financed Pension Systems," *Journal of Institutional and Theoretical Economics*, 1989, *145* (4), 643–658.

Bureau of Economic Analysis, "Survey of Current Business, August 2011: NIPA Tables 3.2 and 3.12," U.S. Department of Commerce 2011.

Cagetti, Marco and Mariacristina De Nardi, "Entrepreneurship, Frictions, and Wealth," *Journal of Political Economy*, 2006, *114* (5), 835–870.

Centers for Medicare and Medicaid Services, "National Health Expenditure Accounts: Age and Gender Tables," Centers for Medicare and Medicaid Services 2012.

Chang, Yongsung, João F. Gomes, and Frank Schorfheide, "Learning-by-Doing as a Propagation Mechanism," *American Economic Review*, 2002, *92* (5), 1498–1520.

Conesa, Juan Carlos and Carlos Garriga, "Optimal Fiscal Policy in the Design of Social Security Reforms," *International Economic Review*, 2008, *49* (1), 291–318.

_ **and Dirk Krueger**, "Social Security Reform with Heterogeneous Agents," *Review of Economic Dynamics*, 1999, *2* (4), 757–795.

De Nardi, Mariacristina, "Wealth Inequality and Intergenerational Links," *Review of Economic Studies*, 2004, *71* (3), 743–768.

_ **, Eric French, and John B. Jones**, "Why Do the Elderly Save? The Role of Medical Expenses," *Journal of Political Economy*, 2010, *118* (1), 39–75.

_ **, Selahattin İmrohoroğlu, and Thomas J. Sargent**, "Projected U.S. Demographics and Social Security," *Review of Economic Dynamics*, 1999, *2* (3), 575–615.

Diamond, Peter A., "Insulation of Pensions from Political Risk," in Salvador Valdés-Prieto, ed., *The Economics of Pensions: Principles, Policies, and International Experience*, Cambridge: Cambridge University Press, 1997, pp. 33–57.

Eggertsson, Gauti B., Neil R. Mehrotra, and Jacob A. Robbins, "A Model of Secular Stagnation: Theory and Quantitative Evaluation," *American Economic Journal: Macroeconomics*, 2019, *11* (1), 1–48.

Fan, Xiaodong, Ananth Seshadri, and Christopher R. Taber, "Estimation of a Life-Cycle Model with Human Capital, Labor Supply, and Retirement," *Journal of Political Economy*, 2024, *132* (1), 48–95.

Geanakoplos, John, Olivia S. Mitchell, and Stephen P. Zeldes, "Would a Privatized Social Security System Really Pay a Higher Rate of Return?," in R. Douglas Arnold, Michael J. Graetz, and Alicia H. Munnell, eds., *Framing the Social Security Debate: Values, Politics, and Economics*, Washington, DC: Brookings Institution Press, 1998, pp. 137–157.

Hansen, Gary D. and Selahattin İmrohoroğlu, "Fiscal Reform and Government Debt in Japan: A Neoclassical Perspective," *Review of Economic Dynamics*, 2016, *21*, 201–224.

Holzmann, Robert, Robert Palacios, and Asta Zviniene, "Implicit Pension Debt: Issues, Measurement and Scope in International Perspective," Social Protection Discussion Paper 0403, World Bank 2004.

Hosseini, Roozbeh and Ali Shourideh, "Retirement Financing: An Optimal Reform Approach," *Econometrica*, 2019, *87* (4), 1205–1265.

Huang, He, Selahattin İmrohoroğlu, and Thomas J. Sargent, "Two Computations to Fund Social Security," *Macroeconomic Dynamics*, 1997, *1* (1), 7–44.

Hubbard, R. Glenn, Jonathan Skinner, and Stephen P. Zeldes, "Precautionary Saving and Social Insurance," *Journal of Political Economy*, 1995, *103* (2), 360–399.

Huggett, Mark, "Wealth Distribution in Life-Cycle Economies," *Journal of Monetary Economics*, 1996, *38* (3), 469–494.

_ **and Gustavo Ventura**, "On the Distributional Effects of Social Security Reform," *Review of Economic Dynamics*, 1999, *2* (3), 498–531.

Imai, Susumu and Michael P. Keane, "Intertemporal Labor Supply and Human Capital Accumulation," *International Economic Review*, 2004, *45* (2), 601–641.

İmrohoroğlu, Ayşe, Selahattin İmrohoroğlu, and Douglas H. Joines, "A Life Cycle Analysis of Social Security," *Economic Theory*, 1995, *6* (1), 83–114.

İmrohoroğlu, Selahattin, Sagiri Kitao, and Tomoaki Yamada, "Achieving Fiscal Balance in Japan," *International Economic Review*, 2016, *57* (1), 117–154.

Kitao, Sagiri, "Sustainable Social Security: Four Options," *Review of Economic Dynamics*, 2014, *17* (4), 756–779.

McGrattan, Ellen R. and Edward C. Prescott, "On Financing Retirement with an Aging Population," *Quantitative Economics*, 2017, *8* (1), 75–115.

Medicare Trustees, "2025 Annual Report of the Boards of Trustees of the Federal Hospital Insurance and Federal Supplementary Medical Insurance Trust Funds," Annual Report, Centers for Medicare and Medicaid Services, Washington, DC 2025.

Nishiyama, Shinichi and Kent Smetters, "Consumption Taxes and Economic Efficiency with Idiosyncratic Wage Shocks," *Journal of Political Economy*, 2005, *113* (5), 1088–1115.

_ **and** _ , "Does Social Security Privatization Produce Efficiency Gains?," *Quarterly Journal of Economics*, 2007, *122* (4), 1677–1719.

Peterman, William B., "The Effect of Endogenous Human Capital Accumulation on Optimal Taxation," *Review of Economic Dynamics*, 2016, *21*, 46–71.

_ **and Erick Sager**, "Optimal Public Debt with Life Cycle Motives," *American Economic Journal: Macroeconomics*, 2022, *14* (4), 404–437.

Ríos-Rull, José-Víctor, "Life-Cycle Economies and Aggregate Fluctuations," *Review of Economic Studies*, 1996, *63* (3), 465–489.

Samuelson, Paul A., "An Exact Consumption-Loan Model of Interest with or without the Social Contrivance of Money," *Journal of Political Economy*, 1958, *66* (6), 467–482.

Sinn, Hans-Werner, "Why a Funded Pension System Is Useful and Why It Is Not Useful," *International Tax and Public Finance*, 2000, *7*, 389–410.

Social Security Administration, *Annual Statistical Supplement to the Social Security Bulletin, 2010*, Washington, DC: Social Security Administration, 2010.

_ , *Annual Statistical Supplement to the Social Security Bulletin, 2011*, Washington, DC: Social Security Administration, 2011.

Social Security Trustees, "The 2011 Annual Report of the Board of Trustees of the Federal Old-Age and Survivors Insurance and Federal Disability Insurance Trust Funds," Annual Report, Social Security Administration, Washington, DC 2011.

_ , "The 2025 Annual Report of the Board of Trustees of the Federal Old-Age and Survivors Insurance and Federal Disability Insurance Trust Funds," Annual Report, Social Security Administration, Washington, DC 2025.

_ , "The 2026 Annual Report of the Board of Trustees of the Federal Old-Age and Survivors Insurance and Federal Disability Insurance Trust Funds," Annual Report, Social Security Administration, Washington, DC 2026.

Tabellini, Guido, "The Politics of Intergenerational Redistribution," *Journal of Political Economy*, 1991, *99* (2), 335–357.

Wallenius, Johanna, "Human Capital Accumulation and the Intertemporal Elasticity of Substitution of Labor: How Large Is the Bias?," *Review of Economic Dynamics*, 2011, *14* (4), 577–591.

Appendix

Reforming Social Security Without a Losing Generation:
A Pareto-Improving Transition under a Debt Ceiling

HongSeok Kim

Contents

A Model and calibration details

This appendix supports the model in Section 2 and the calibration and welfare measures in Section 3.

A.1 Demographic projections

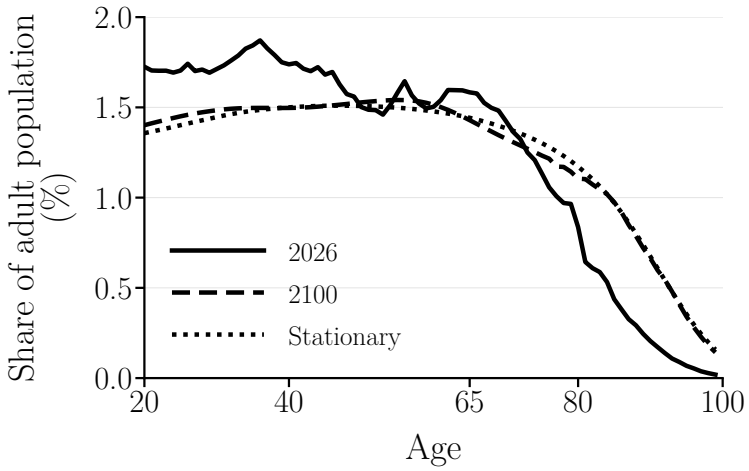


Figure A.1: Projected age distribution of the adult population

Notes. Share of the population aged 20–99 at each age, in percent. The 2026 and 2100 distributions are from the intermediate projection of the 2026 Trustees Report. The stationary distribution holds births at their 2100 level and applies the 2099–2100 net survival and migration factors at every age, separately by sex.

A.2 Calibration

I normalize productivity to $A_{2010} = 1$ and set L_0 to the effective labor households supply in 2010. With a capital–output ratio of three, initial output is $Y_0 = 3^{\alpha/(1-\alpha)} L_0$. To express model values in dollars, I scale output so that Y_0 equals 2010 GDP of \$15,049 billion (Social Security Trustees, 2025), using the factor $D = \$15{,}049 \text{ billion}/Y_0$. Per-person values divide by adults aged 20–64 in 2010. The initial stocks are

- Capital is $K_{2010} = 3Y_0$, or \$45.147 trillion.
- Public debt is $B_{2010} = 0.5298\, Y_0$. This value uses federal debt held by the public on December 2009: \$7,811 billion (Treasury). I convert this amount to 2010 dollars and divide it by 2010 GDP.

- Trust-fund reserves are $F_{2010} = 0.1723\,Y_0$. OASDI assets in 2009 were \$2,540 billion in nominal dollars, or \$2,592 billion in 2010 dollars (Social Security Trustees, 2011, Table III.A3).

- Household wealth is $K_{2010} + B_{2010} = 3.530\,Y_0$. I scale the initial asset distribution to match this total.

Table A.1: Parameters set outside the model

Parameter	Value	Description and source or basis
Preferences and technology		
σ	2	Utility curvature. Sensitivity analysis with $\sigma = 4$
b_*	15,004	Bequest scale (2010 dollars). Average retired-worker benefit in payment in 2010, after the scaling to 2010 OASDI outlays (Social Security Administration, 2010; Social Security Trustees, 2011)
g	0.0113	Productivity growth. Real-wage growth in the Trustees' intermediate projection (Social Security Trustees, 2025)
α	1/3	Capital share. —
δ	0.0661	Capital depreciation. $K/Y = 3$ (De Nardi et al., 1999; Cagetti and De Nardi, 2006) and a 4.5% pre-tax return
γ	0.05	Skill exponent in learning by doing. Fan et al. (2024)
Social Security		
ρ	0.4826	Retirement benefit at 65 / taxable earnings. 2010 program accounts (Social Security Administration, 2010, 2011)
ζ_{2010}	0.512	Effective taxable-payroll share, using net contributions. Social Security Trustees (2011). Future path from the 2026 Trustees Report
φ_s	0.929	Benefit adjustment for birth cohorts from 1960 onward. Statutory benefit reduction at age 65
θ	0.025	Income-tax rate on benefits, credited to the trust fund. Model setting. The 2010 average rate was 3.4%
Taxes, public debt, and the consumption floor		
τ^G	0.246	Earnings-tax rate. Balances the 2010 reference budget
τ^k	0.30	Capital-income tax rate. Kitao (2014)
$\bar{b}$	1.00	Debt ceiling / GDP. CBO projection: debt exceeds 100% of GDP around 2030
$\underline{c}_0$	0.097	Floor / 2010 GDP per adult aged 20–89. 2010 SSI benefit relative to the average retirement benefit (Social Security Administration, 2011)
Health spending		
p_j^M	0.594 / 0.083	Medicare share: ages 65+ / under 65. NHEA tables by age and payer (Centers for Medicare and Medicaid Services, 2012), scaled to 2010 spending
p_j^O	0.14 / 0.22	Out-of-pocket share: ages 65–84 / 85+. NHEA tables by age and payer (Centers for Medicare and Medicaid Services, 2012)
$\mathcal{M}_t$	3.45 / 6.23 / 6.69	Medicare / GDP (%): 2010 / 2050 / 2100. Medicare Trustees (2025)

A.3 Parameters set outside the model

Preferences and technology. Table A.1 lists the values set outside the model. Depreciation satisfies $\delta = \alpha/3 - 0.045 = 0.0661$. This value gives an after-tax return of about 3%. The bequest scale $b_* = \$15{,}004$ is the count-weighted average retired-worker benefit in payment at the start of 2010: the 2009 average monthly benefits by single age, annualized ($13,946 on average) and multiplied by the factor 1.0759 that matches benefits in payment to 2010 OASDI outlays, as described below.

The skill exponent $\gamma = 0.05$ comes from Fan et al. (2024, Table 3). Their Ben-Porath investment technology differs from (2). Thus, I use this value as a calibration benchmark. It is not a direct estimate of learning by doing.

The retirement-benefit coefficient. The initial retired-worker benefit at age 65 is $\hat{b}_s = \varphi_s \rho\, \bar{e}_{s-1}$, where $\bar{e}$ is average taxable earnings per covered worker. I calculate ρ from the data:

$$\rho = \frac{12 \times \$1{,}370}{\$34{,}065} = 0.4826.$$

The numerator is twelve times the average monthly retirement benefit for workers newly entitled at age 65 in 2010. It uses December 2010 benefit rates (Social Security Administration, 2011, Table 6.A4). The denominator is 2009 taxable earnings per covered worker, converted to 2010 dollars. Taxable earnings were $\$5,271$ billion across 158 million workers, or $\$33,375$ per worker (SSA Annual Statistical Supplement, Table 4.B11). I convert this amount using the CPI ratio 213.967/209.630.

The model has no taxable maximum or covered-worker count, so I link $\bar{e}_t$ to the model's wage bill per working-age adult:

$$\bar{e}_t = \psi \frac{w_t L_t}{n_t^W}, \qquad \psi = \frac{\text{2010 taxable payroll per covered worker}}{(1-\alpha) \times \text{2010 GDP per adult aged 20–64}}.$$

The denominator of ψ is the 2010 labor income per working-age adult implied by the model's labor share. So ψ converts the model's wage measure into taxable earnings per covered worker in 2010. I hold ψ fixed after 2010, so benefits at age 65 grow one-for-one with the wage bill per working-age adult.

The normal-retirement-age adjustment $\varphi = 0.867/0.933 = 0.929$ applies to cohorts born in 1960 or later. For these cohorts, the normal retirement age is 67. A claim at age 65 pays 86.7% of the primary insurance amount. For the cohorts used to set ρ, the normal retirement age is 66, and a claim at age 65 pays 93.3%. I omit the phase-in for the 1955–1959 birth

cohorts.

Initial retirement benefits and coverage. Retirement benefits in payment in 2010 use the 2009 distribution of retired-worker beneficiaries and their average monthly benefits. The data cover single ages from 62 to 89 (Social Security Administration, 2010, Table 5.A1.1). I split each group's count across single ages in proportion to each age's population.

I multiply all benefit amounts by a common factor of 1.0759. This adjustment includes disability, survivor, and auxiliary benefits. It makes annualized 2009 amounts equal 2010 OASDI benefit outlays less lump-sum death payments: $701 billion (Social Security Trustees, 2011, Table III.A3). The parameter ν is the share of adults aged 65 and over who receive a retired-worker benefit in 2010. Later changes in this share enter through ξ_t, the number of retired-worker beneficiaries per person aged 65 and over relative to 2010 (Appendix A.4).

Payroll coverage. Payroll tax applies only to earnings below the taxable maximum, which was $106,800 in 2010 and $184,500 in 2026, and only to covered work. The taxable share ζ_t captures both limits. I compute it from projected payroll receipts $\mathcal{P}_t$:

$$\zeta_t = \frac{\tau_t^{ss}}{0.124}, \qquad \tau_t^{ss} = \frac{\mathcal{P}_t}{(1-\alpha)Y_t},$$

where τ_t^{ss} is the effective payroll tax rate on labor compensation. In 2010, net payroll contributions were $637.283 billion. The model uses these contributions to set $\tau_{2010}^{ss} = 0.0635$ and $\zeta_{2010} = 0.512$. The one-time general-fund reimbursement of $2.4 billion enters separately. The retained SSA comparison series includes the rounded reimbursement in its initial payroll benchmark, implying $\zeta_{2010} = 0.514$. The resulting model-to-SSA payroll ratio is about 0.9962 throughout the projected path.

From 2026 to 2100, I calculate τ_t^{ss} as 0.124 times the Trustees' projected taxable-payroll-to-GDP ratio, divided by $1-\alpha$. I interpolate linearly between the reported values for 2026, 2034, 2050, 2060, 2085, and 2100 (Social Security Trustees, 2026, Tables IV.B3 and VI.G1). Between 2010 and 2026, I interpolate linearly between the 2010 and 2026 values.

I impose this path relative to the projected number of covered workers. Thus, the model's payroll receipts relative to labor compensation match the projected ratio in every year. The path remains constant after 2100.

Income taxes and the debt ceiling. The capital-income tax applies to $r(K + B)$. The earnings-tax rate balances the 2010 reference budget:

$$\tau^G = \frac{G^{\text{purch}}_{2010} + S^G_{2010} + (r_{2010} - g)B_{2010} - \tau^k r_{2010}(K_{2010} + B_{2010})}{w_{2010}L_{2010}} = 0.246.$$

Purchases equal 2010 federal consumption and gross investment, less consumption of fixed capital and Social Security administration. They total \$1,086 billion, or 7.22% of GDP.

Transfers outside Social Security equal total federal expenditures less interest, Social Security outlays, and purchases. They total \$1,828 billion, or 12.15% of GDP (Bureau of Economic Analysis, 2011, Tables 3.2 and 3.12). I replace interest payments from the data with the model's required primary surplus on a balanced-growth path, $(r - g)B$.

I allocate transfers by age. Medicare and other retirement programs go to adults aged 65 and over. Unemployment insurance, workers' compensation, nutrition assistance, and refundable credits go to adults aged 20–64. I divide the remaining transfers equally among all adults. With the medical block, I remove the 2010 Medicare amount from cash transfers and provide it in kind. The debt ceiling is $\bar{b} = 1$.

Consumption floor. The consumption floor applies from age 65. The ratio of the 2010 federal Supplemental Security Income benefit for an individual to the average retired-worker benefit provides its benchmark: \$8,088/\$13,968 = 0.579 (Social Security Administration, 2011). The implemented floor is 9.7% of 2010 GDP per adult aged 20–89. It grows with productivity.

Medical spending. The National Health Expenditure Accounts provide personal health care spending per person by age (Centers for Medicare and Medicaid Services, 2012). In 2010, spending was \$3,628 at ages 0–18 and \$4,422 at ages 19–44. It was \$8,370 at ages 45–64, \$16,872 at ages 65–84, and \$32,411 at ages 85 and over.

Before scaling, Medicare pays 0.07 of spending below age 65 and 0.50 at ages 65 and over. I multiply both shares by a common factor to match Medicare spending on the 2010 population to 2010 Medicare outlays. These outlays were \$518 billion, or 3.4% of GDP. The scaled shares are 0.083 and 0.594. Households pay out of pocket a share of personal health care spending equal to 0.11 at ages 0–18, 0.14 at 19–44, 0.13 at 45–64, 0.14 at 65–84, and 0.22 at 85 and over.

The Medicare path $\mathcal{M}_t$ is the sum of Part B, Part D, and Hospital Insurance spending relative to GDP. For Hospital Insurance, I multiply its cost rate by its taxable-payroll-to-GDP

ratio. This ratio is 0.4355 in 2024. I scale it using the projected path of OASDI taxable payroll relative to GDP (Medicare Trustees, 2025, Tables III.B7, III.C10, III.D6).

A.4 Projected paths taken as data

Each follows the Trustees' projections from 2010 to 2100 and stays at its 2100 value afterward.

- *Retired-worker wedge* ξ_t: retired-worker beneficiaries per person aged 65 and over, relative to 2010 (Social Security Trustees, 2025, Table V.C4). It equals 1 in 2010, 0.997 in 2034, 1.012 in 2050, and 1.044 in 2100.
- *Other benefits* $\mathcal{H}_t^O$: disability, spouse, child, and survivor benefits. They equal 1.5% of GDP in 2010 and keep their projected share of scheduled cost (Social Security Trustees, 2025, Table IV.B1). Their age allocation stays at its 2010 pattern.
- *Trust-fund return* R_t^F: the gross real return on trust-fund reserves. It uses the historical effective yield through 2025 and the Trustees' projected yields from 2026 (Table VI.G1), deflated by the Trustees' CPI.
- *Fund outlays* $\mathcal{O}_t$: administration and vocational rehabilitation ($6.58 billion in 2010) grow with the population. The railroad-retirement interchange ($4.39 billion) grows with the population aged 65 and over (Social Security Trustees, 2011, Table III.A3).
- *Medicare* $\mathcal{M}_t$: 3.45% of GDP in 2010, 6.23% in 2050, and 6.69% in 2100.

A.5 Heterogeneity, estates, and wages

Wealth. The 2010 Survey of Consumer Finances (SCF) provides the initial asset dispersion and the estate target. I pool respondents and spouses aged 20–89. When a spouse is present, I divide family net worth equally between spouses. I set non-positive net worth to zero because households cannot borrow. The source survey contains 6,482 families. I pool its five imputations and divide survey weights by five. Figure A.2 plots three measures by five-year age group: the share with zero wealth, the standard deviation of log positive wealth, and the share of inheritances received in 2010 by recipient age. I divide each five-year inheritance share equally among its single ages.



Figure A.2: Distributional inputs from the 2010 SCF

Notes. Values by five-year age group, plotted at the group midpoint. Left: share of adults with zero net worth. Middle: standard deviation of log net worth among adults with positive net worth. Right: share of inheritances received in 2010, by recipient age. Respondents and spouses aged 20–89 are pooled, family net worth is divided equally between spouses, and non-positive net worth is set to zero.

Initial asset levels. Initial assets by age take the age profile of the benchmark steady state (the reference economy), $\bar{a}_j^{\text{ref}}$, and scale it so that total household assets in 2010 equal capital plus government debt:

$$\bar{a}_{j,2010} = \lambda \bar{a}_j^{\text{ref}}, \qquad \lambda = \frac{K_{2010} + B_{2010}}{\sum_{j=20}^{89} n_{j,2010} \bar{a}_j^{\text{ref}}}.$$

The scaling uses the actual 2010 population and changes only the level, not the shape, of the profile. Initial estates are zero. This scaling is separate from the one applied to SCF wealth for the estate target.

The estate target. Estates are the wealth of people who die. Let n_a be the 2010 population at age a, q_a the SSA death probability, and $\bar{W}_a$ average SCF wealth per adult at that age. The target is

$$\frac{\text{estates}}{\text{GDP}} = \underbrace{\frac{\sum_a n_a q_a \bar{W}_a}{\sum_a n_a \bar{W}_a}}_{\text{share of wealth left by decedents}} \times \underbrace{\frac{K_{2010} + B_{2010}}{Y_{2010}}}_{\text{model wealth / GDP}} = 0.01524 \times 3.530 = 5.378\%.$$

I compute the same statistic in the reference economy with 2010 population weights.

Wages and skill. I use the Current Population Survey (CPS) outgoing rotation groups from January through December 2010. The sample includes workers aged 20–64 with positive hours and earnings. I calculate hourly wages as weekly earnings divided by usual weekly hours.

Average hourly wages by five-year age group give two wage-ratio targets. The ratio for ages 40–44 relative to ages 20–24 is \$23.20/\$12.18 = 1.905. The ratio for ages 60–64 relative to ages 40–44 is \$23.37/\$23.20 = 1.007. The standard deviation of log hourly wages for workers aged 20–24 is 0.493. I calculate it using CPS earnings weights and 13,746 observations. I use this value for entrants and for the within-age skill dispersion imposed on the other opening cohorts. The main text reports it as $s_h = 0.493$.

The data measure wage dispersion. I assume that this dispersion represents skill dispersion. I also assume that skills follow a lognormal distribution. I assume independent asset and skill draws because the SCF and CPS samples are not linked, and approximate normal draws with nine Gauss–Legendre nodes per dimension on $[-4, 4]$. I normalize the approximation to preserve averages.